ETH
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LRE Language Reasoning Education

Automatic Evaluation of Tutor Responses Using Large Language Models

An Analysis of Pedagogical Effectiveness

Bachelor's Thesis

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Chapter 1

Abstract

One of the significant challenges in the research and development of AI-based educational tools is the lack of standardized evaluation methods for assessing pedagogical performance. While various approaches exist to recognize good pedagogy, a criteria-based approach defines rubrics that a tutor must meet to ensure high-quality teaching. Building on the foundations of previous studies, this thesis aims to analyze the capability of state-of-the-art large language models (LLMs) to evaluate tutor responses in dialogue-based tutoring. Specifically, we propose an automatic evaluation framework that uses LLMs to classify whether a tutor fulfills specific pedagogical rubrics, which are inspired by Jurenka et al. [10] and other pedagogical literature.

Our research involves testing these models on diverse datasets, including real-life student-teacher dialogues and synthetic conversations. First, we manually annotate the datasets to create a ground truth for each rubric. Then, we prompt language models of varying sizes and methodologies to perform the same task. We evaluate the performance of the models by comparing their responses to human annotations. By identifying the most accurate method, we propose an automatic evaluation framework and demonstrate its application in a case study. Finally, we explore the potential of fine-tuning models. Since the ground truth dataset is very small for fine-tuning, we perform data augmentation by generating data using a large language model and use this augmented data to train a smaller classification model.

Our findings show that, in general, few-shot prompting with detailed system prompts—where the classification scenarios are explicitly listed—works best. Larger models tend to perform better overall; however, the specific model that performs best varies depending on the rubric. The variation in performance between rubrics was primarily due to imbalances in the ground truth datasets and the definition and phrasing of the rubrics. The user study indicated that for some rubrics, practical application worked very well, while for others, it was more mediocre. In all cases, models performed best for the rubric "Promote Active Engagement". In the fine-tuning approach, although the classification model performed poorly, we made a secondary finding that fine-tuning the large language model performed exceptionally well, even with the small size of the training set.

Our findings contribute to the ongoing discourse on evaluating AI-powered tutors and provide valuable insights into the most effective methodologies for enhancing pedagogical interactions in AI-driven learning environments. Our dataset and code can be find at:

Acknowledgements

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Chapter 2

Introduction

2.1 Background

Since the 1950s, artificial intelligence (AI) has been a topic of great interest, demonstrating significant advancements over the decades. Among many others, the development of generative AI marked a groundbreaking shift. Since the release of GPT-3 by OpenAI in 2020, AI has become deeply integrated into our daily lives like never before. Alongside personal adoption and industries transforming their businesses to align with a rapidly changing world, the emergence of generative AI has sparked new fields of research, exploring its potential in previously uncharted areas.

AI is no longer limited to tasks such as playing chess or Go—it can now engage in conversations, create art, and perform tasks that imitate human qualities such as friendliness, creativity, and reasoning. This remarkable human-likeness has allowed AI systems to assume roles traditionally filled by humans. Many individuals and organizations have embraced this new world. People often report increased productivity when incorporating AI into their workflows, while industries such as healthcare, marketing, and education can offer more personalized services to their customers without compromising—and often even improving—efficiency. [7]

However, this rapid adoption has also raised important ethical and practical questions. Can these systems be trusted? Are they fair and unbiased? Concerns regarding privacy, data security, and lack of control widespread. [7]

2.2 AI in Education

Similarly, questions about the role of AI in education have sparked debates. While AI tutors have the potential to democratize education by making personalized learning accessible to everyone, educators and learners alike express concerns about privacy and the legitimacy of AI as a teaching tool. Despite these concerns, research in this field continues to grow, and AI is already being used to support teachers and learners in various ways. [10]

In a world where AI-driven teaching systems are being developed and the role of human teachers is being questioned, it is crucial to establish a method to evaluate their pedagogical abilities. Pedagogy is inherently complex, time-consuming, and expensive to assess manually. In this study, we explore the possibility of automating this process.

2.3 Focus of this Work

To achieve this, we evaluate the ability of generative AI, particularly large language models (LLMs) to assess pedagogy. By conducting a side-by-side comparison of human evaluations and AI evaluations of tutors' pedagogical skills, we aim to measure how accurately large language models can replicate human judgment. We experiment with different approaches to optimize AI performance and report our findings.

2.4 Thesis Organization

The structure of this thesis is organized as follows:

Chapter 2: Related Works This chapter provides an overview of the research landscape on large language models (LLMs) and their use in education. We identify key challenges and gaps in the field, emphasizing the significance of our study. Finally, we introduce our proposed solution.

Chapter 3: Methodology This chapter defines the scope of our task, specifying which rubrics the LLMs are designed to measure and which datasets they will be tested on. We outline the methods used to collect human annotations, which serve as ground truth data for evaluating the performance of the LLMs. Additionally, we assess the quality of the collected annotations and describe the processes involved in preparing the data for subsequent analysis.

Chapter 4: Experiments and Results In this chapter, we describe how the LLMs were tasked to perform the defined objectives and outline the different approaches used. We compare different methods and their performance using statistical evaluation against the ground truth data. Based on these results, we propose the most effective evaluation framework and demonstrate its application in a case study.

Chapter 5: Discussion This chapter provides a theoretical analysis of the results, exploring why certain outcomes were observed and attempting to generalize the findings. Additionally, we offer recommendations for future research in this area.

Chapter 3

Related Work

3.1 The Rise of Large Language Models and its Application

In recent years, advancements in large language models (LLMs) have progressed dramatically [8]. Every year, new and better-performing LLMs are released [8], with many of them being open source [23], [4]. This rapid development has led to various applications of these models across research communities and industries, all seeking to leverage this state-of-the-art technology in diverse fields. Modern large language models excel at understanding and generating human-like text, opening up opportunities to replace or support tasks that traditionally required human input [8].

In this paper, we explore the potential applications of LLMs in education. This chapter reviews key studies on the use of AI in education to date and highlights the existing challenges. We then define the problem we aim to address and present our approach to solving it.

3.2 AI in Dialogue Tutoring

The primary objective currently is to develop AI systems that can act as tutors in one-on-one dialogues. This is considered one of the most general use cases, requiring the integration of multiple educational capabilities into a single system [10].

Macina et al. [15] explored the potential of generative language models for dialog tutoring. They tested various models on two datasets of tutor-student dialogues focused on language learning, using approaches such as providing the model with dialog history, grounding information, and relevant dialog acts, or offering a training corpus from which the model could select responses to reduce inaccuracies. Evaluation using both automated and human methods showed improvements over previous techniques but revealed significant limitations, including inadequate datasets, model performance, and evaluation metrics.

Handa et al. [9] on the other hand, focused on using AI to enhance human tutor responses rather than generating new responses. The study focused on the use of teachers' growth mindset supportive language (GMSL). Observing a lack of coaching resources for teachers in this area, the researchers annotated a dataset of elementary math classroom transcripts, identifying unsupportive utterances and providing human expert-revised GMSL reframings. They also prompted large language models to generate similar reframings. Evaluations by students and teachers showed a preference for both expert-generated and AI-generated reframings over the original utterances, with AI-generated reframings slightly outperforming those created by experts.

Wang et al. (2024) [25] proposed a new approach, introducing "Bridge," a method that uses cognitive task analysis to translate an expert's thought process into a decision making model. Using these decisions, they prompted different large language models and evaluated their responses, comparing them to the responses

without the decisions. The human evaluation demonstrated that embedding expert thought processes in large language models significantly improved both pedagogical outcomes and human preferences.
A recent effort to use AI as a tutor was undertaken by Jurenka et al. (2024) [10]. They tested two methods: prompting Gemini 1.0 and fine-tuning it, creating LearnLM Tutor using five curated datasets. To measure AI tutors' performance, they developed seven educational benchmarks that combined human and automated evaluations. Their study highlighted the importance of standardized pedagogical benchmarks for assessing AI tutors.

3.3 Evaluation of Tutors

A crucial challenge after developing AI to act as a tutor is to evaluate the pedagogical ability of the models that were developed. While human evaluation offers a potential solution, it is costly, time-consuming and difficult to scale [10]. It is therefore crucial to explore the possibilities of automating this process.

3.3.1 Challenges in evaluating pedagogy

A significant challenge in automating this evaluation lies in the lack of universally agreed-upon rubrics defining effective pedagogy. While theories in learning science outline the qualities of effective tutors, a standardized framework to evaluate AI tutors is lacking[10]. Consequently, even though numerous studies have assessed the pedagogical abilities of LLMs, directly comparing results remains difficult [10]. Broadly, two automatic evaluation approaches have emerged to define effective pedagogy.

Comparing AI and Human Tutor Responses

The first approach compares AI-generated responses to those of human tutors. For instance, Macina et al. [15] used metrics such as sBLEU and BERTScore [19] to measure lexical overlap and semantic similarity between AI-generated responses and ground-truth responses. However, their findings revealed low correlations between these metrics and human evaluations, emphasizing the need for improved automatic evaluation methods.

Defining own set of criteria

The second approach involves defining a set of criteria that characterize an ideal tutor, applicable to both AI and human educators. This is the common approach used for both automatic and human evaluation. For example, Handa et al. [9] emphasized the importance of GMSL in evaluating tutor performance. Kakarla et al. [11] proposed an automatic evaluation approach to assess tutor responses to student errors. However, their study was limited by the inability of LLMs to reliably identify student mistakes. Tack and Piech [22] evaluated pedagogical ability using rubrics such as "likely said by a teacher", "understanding the student", and "helping the student" using human evaluation. Human raters ranked different tutor responses for each conversation snippet. While it would be feasible to automate this evaluation framework, the evaluation rubrics do not capture the full richness of pedagogy [10], as it was developed to soley test AI teachers [22].

Combining Both Approaches

Jurenka et al. [10] addressed these limitations by incorporating both human-likeness and criteria-based approaches in their seven educational benchmarks. They highlight the usefulness and importance of automatic evaluations for making rapid model development progress by offering quick insight into pedagogical capabilities [10]. For the criteria-based approach, they developed eight metrics from five pedagogy dimensions in collaboration with education experts to define good pedagogy.

3.3.2 Multi-turn Dialogue Tutoring Datasets

Another significant challenge in this domain is the limited availability of large, open-source datasets capturing tutor-student dialogues. Many studies rely on privately developed datasets constrained by privacy considerations [14]. To address this issue, researchers have developed open datasets while maintaining pedagogical quality.

Generated Datasets

For instance, Stasaski et al. [20] collected tutoring dialogues using crowdworkers role-playing as students and tutors. Each student turn was paired with three responses from distinct tutors, and all dialogue turns were annotated with action types. However, since the crowdworkers were not trained educators, the dataset's pedagogical quality is relatively low. Macina et al. [14] generated tutoring dialogues by pairing human teachers with large language models simulating common student errors. This approach ensured high pedagogical quality but lacked variability, which may be valuable in some contexts.

Human-Human Datasets

Other open datasets, such as Conversational Uptake [6] and TalkMoves [21], consist of transcripts of real elementary math classrooms. However, real-life transcripts often lack situational clarity, as visual and auditory cues are not included. The dataset developed by Wang et al. [25] is relatively small but valuable, as it includes multiple tutor responses to individual student dialogue turns, with varying pedagogical quality. They constructed this from a dataset of 700 real tutoring conversations, which were additionally annotated by experts with their responses. The selection of datasets for this thesis is discussed further in Chapter 3.

3.4 Our Solution

Collecting all the points, this is the problem we aim to address: With the growing focus on developing AI tutors, there is a significant demand for effective methods to evaluate their pedagogical abilities. While human evaluation remains the ideal approach, it is neither cost-effective nor time-efficient.
Therefore, we propose an automatic evaluation method for pedagogical ability, leveraging recent advancements in large language models (LLMs). We envision this as a first step toward the development of a standardized automatic evaluation framework, as no such framework currently exists. Such a framework would be immensely valuable for ongoing research, as it would enable AI tutors to be assessed and compared in a consistent manner.
With this goal in mind, we focus on criteria-based automatic evaluation. While Jurenka et al. [10] propose a solution for defining rubrics for good pedagogy and perform automatic evaluations of their AI tutors, they do not provide an open-source framework. Additionally, their approach to evaluating some rubrics requires supplementary information (e.g., the correct answer to the question). This dependency makes their framework challenging to apply in conversations where such information is unavailable.
To address these limitations, we propose a solution that relies exclusively on open-source datasets and requires no additional data beyond the conversation being evaluated. Our approach is as follows:

- Define the Task and Rubrics: We will first define the task and the pedagogical rubrics we aim to evaluate and identify suitable open-source datasets.
- Human Annotation: The datasets will be annotated by humans, with an emphasis on ensuring high annotation quality to create reliable ground-truth data for each rubric. To further support ongoing research in this area, we will also open-source this dataset.

- Model Evaluation and Comparison: We will experiment with various models and approaches, comparing their performance using the human annotation to gain insights that can help optimize the evaluation framework.

With this solution, we believe it represents the first step toward creating a consistent and accessible framework that others can use. Furthermore, this framework will provide insights into the capabilities of large language models in assessing human qualities, highlighting their strengths and limitations.

Chapter 4

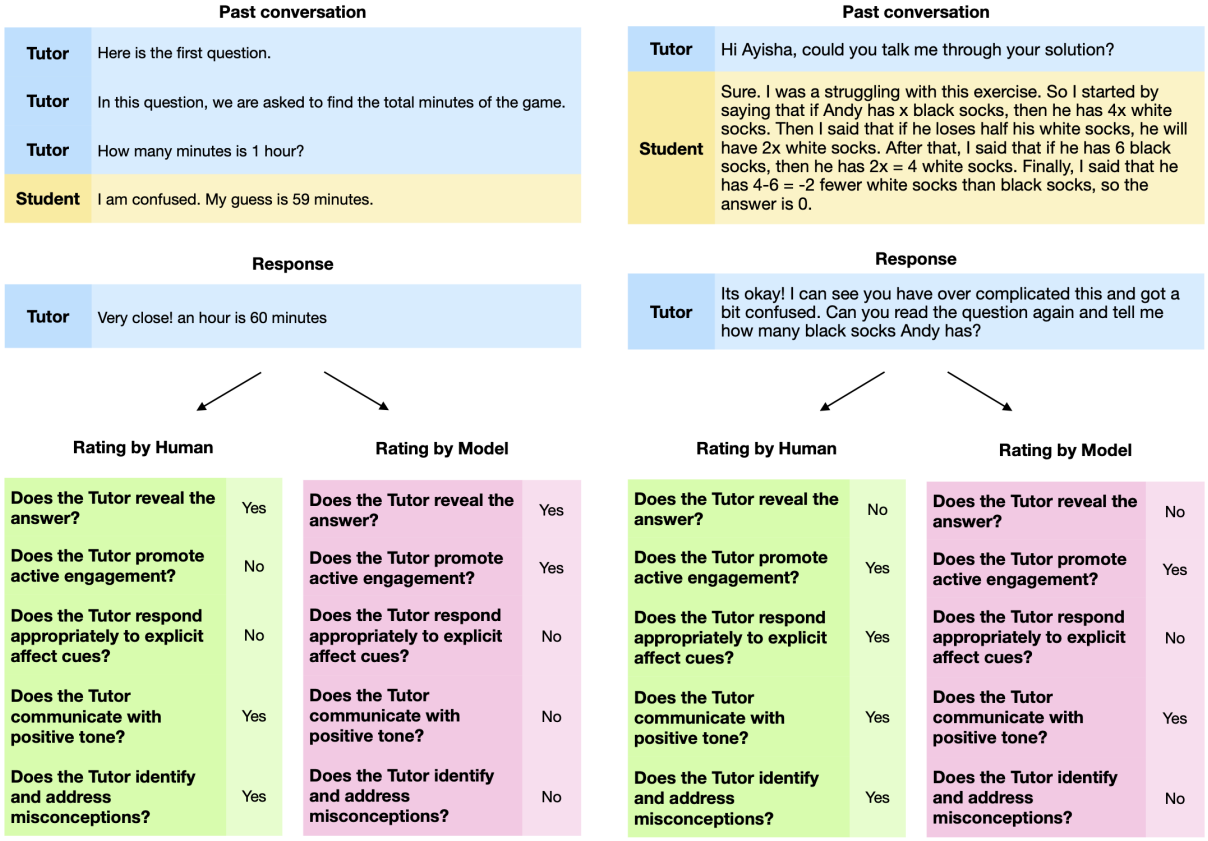
Methodology

4.1 Task Definition

We first define our main task for this study.
We limit our analysis to 1:1 conversational tutoring scenarios. In evaluating pedagogical quality, our focus is on the quality of individual tutor utterances rather than entire conversations. Specifically, the task involves assessing a tutor's response in the context of preceding dialogue between the student and tutor. This approach simplifies the task by targeting the pedagogical quality of each response instead of evaluating the performance throughout the entire dialogue. Furthermore, a consistently high-quality response indicates overall good performance, whereas a well-conducted conversation does not necessarily imply consistent quality in each response.
The tutor response in each conversation will be tested based on selected rubrics first by humans and then by Large Language Models (LLMs). We will assess the ability of Large Language Models (LLMs) to evaluate effective pedagogy by comparing their ratings with those of humans to analyze how accurately they can rate each rubric. An overview of this objective is shown in Figure 4.1.

4.2 Rubric selection

We chose specific rubrics to test for our criteria-based automatic evaluation. As mentioned in Chapter 3, our goal is to build upon the proposed solution from Jurenka et al. [10]. Each high dimensional pedagogy principle "Encourage active learning" [2], [24], [26], [16], "Motivate and stimulate curiosity" [18], [12] and "Deepen metacognition" [5], [3], [13], were inspired and chosen from various education studies. Jurenka et al. [10] then translated them into different measurable rubrics. We selected a subset of the rubrics they presented in their solution. Our choice of rubrics was guided by the feasibility of applying them to our datasets and the scope of this study. For each rubric, we gained a deeper understanding of its precise meaning through the prompts and examples used to guide models in automatic evaluation [10]. This understanding informed the development of instructions for data annotation, as well as the prompts employed in the prompting approach, which will be introduced in subsequent sections. The selected rubrics and their corresponding pedagogy dimensions are shown in Table 4.1. In Table 4.2, the numerical identifiers for each rubric are provided, which we will use to reference them throughout this paper.
For rubrics "Do not reveal the answer", "Respond appropriately to explicit affect cues" and "Identify and address misconceptions", it is important to notice that these rubrics cannot be applied in every situation. For the tutor to not reveal the answer, the student must be currently trying to solve a question. For the tutor to respond appropriately to explicit affect cues, the student must show explicit affect cues. Lastly, for the tutor to identify and address misconceptions, the student must have and demonstrate their misconceptions.



Evaluation

Rubric	Error Rate	F1 Score
Does the Tutor reveal the answer?	0	1
Does the Tutor promote active engagement?	0.5	0.67
Does the Tutor respond appropriately to explicit affect cues?	0.5	0
Does the Tutor communicate with positive tone?	0.5	0.67
Does the Tutor identify and address misconceptions?	1	0

Figure 4.1: Overview of task

Encourage active learning	Do not reveal the answer Promote active engagement
Motivate and stimulate curiosity	Respond appropriately to explicit affect cues Communicate with positive tone
Deepen metacognition	Identify and address misconceptions

Table 4.1: Selected Rubrics

1	Do not reveal the answer
2	Promote active engagement
3	Respond appropriately to explicit affect cues
4	Communicate with positive tone
5	Identify and address misconceptions

Table 4.2: Numerical Identifier for each Rubric

Therefore, for testing the tutor responses for these rubrics, we have to differentiate conversations where these apply.

4.3 Dataset selection

After defining the task, we searched for suitable datasets. An ideal dataset would contain 1:1 tutor-student dialogues illustrating both strong and weak pedagogical performance, with tutor responses detailed enough for evaluation. We selected two datasets that complement each other well for our objectives: the Bridge dataset developed by Wang et al. [25] and the MathDial dataset developed by Macina et al. [14].
The Bridge dataset contains 700 real tutoring conversations, including the conversation history, the tutor's original response, and an expert's pedagogically improved response [25]. While this dataset was suitable for showing contrasting pedagogical quality, it lacked the content exchanged via the blackboard used during sessions and sometimes included typographical errors and unpredictable student behaviors.
MathDial comprises tutoring dialogues generated through the interaction of human tutors with LLMs that simulate common student errors [14]. This dataset is rich in high-quality pedagogical examples, consisting of over 3,000 conversations. However, it rarely includes examples of poor pedagogy. Additionally, unlike the Bridge dataset, MathDial does not provide single-out responses, requiring us to truncate conversations up to the second tutor utterance. We defined the second tutor utterance to be the tutor response to be evaluated. This truncation was necessary as the first tutor utterances are often introductory, and subsequent utterances lack predictable patterns for systematic filtering.

4.4 Data annotation

We established a ground truth data set through human annotation, which will be used to test and fine-tune the LLMs. We randomly selected 150 conversations from each dataset, Bridge and MathDial. A web application was developed for annotation purposes and 15 annotators, all pursuing bachelor's or master's degrees, were instructed on the rating criteria before they began. The annotators were aged 19 to 25, 8 were women and 7 were men. All resided in Switzerland or Germany. Each conversation was reviewed by exactly two annotators, who evaluated past dialogues and current tutor responses based on five distinct rubrics. Annotators could log into the web interface, which displayed the number of conversations they had rated.

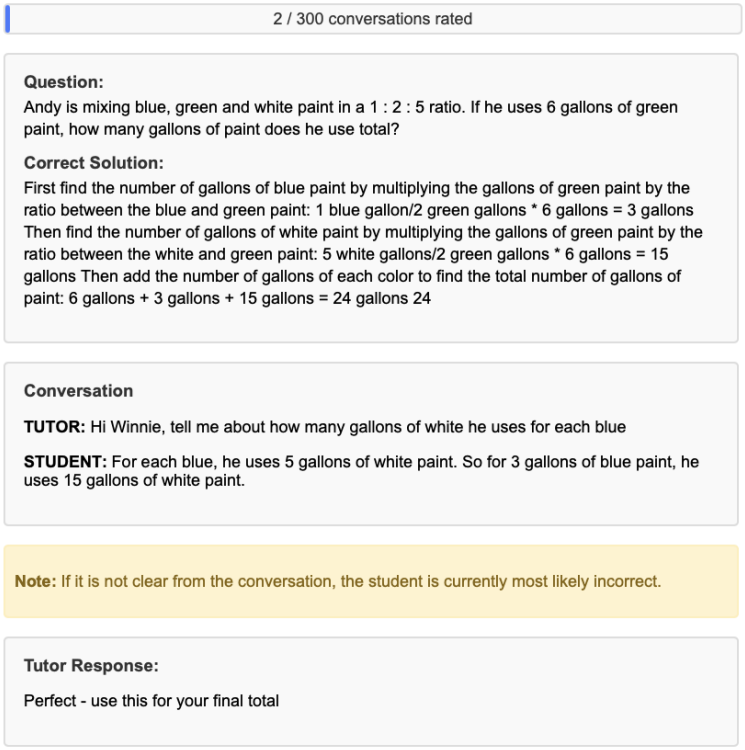


Figure 4.2: Conversation from MathDial dataset

The interface, as shown in Figures 4.2, 4.3, 4.4, 4.5 featured a conversation box that displayed the ongoing dialogue between the student and the tutor, with additional information specific to the dataset. The additional information and examples shown for each rubric is shown in detail in Appendix A. Below the conversation box, there was a question for each rubric. Each rubric included examples and explanatory information. Annotators could choose "Yes" or "No" for the rubrics, with "Not Applicable" available for certain rubrics, as explained in Section 4.2. For conversations from the MathDial dataset, the question the student is currently trying to answer and the correct solution were additionally shown. For conversations from the Bridge dataset, the current lesson topic was displayed.

4.5 Analysis of data annotation

We then analyzed the collected data annotations. First we analyze the number of conversations each annotator annotated as shown in Table 4.3.

4.5.1 Annotation Distribution and Dataset Imbalance

Since user 01 was the only annotator who reviewed all 300 conversations, we visualized their annotations as a representative sample to gain an initial understanding of the annotation distribution. This approach allowed us to assess whether the dataset was balanced, particularly in terms of the number of "Yes" and "No" annotations compared to "Not Applicable" annotations for each rubric. A high prevalence of "Not Applicable" annotations could be problematic, as conversations marked this way would be excluded from further analysis of the rubric.
As illustrated in Figure 4.6, there was a noticeable imbalance in the annotations for Rubric 3 ("Respond

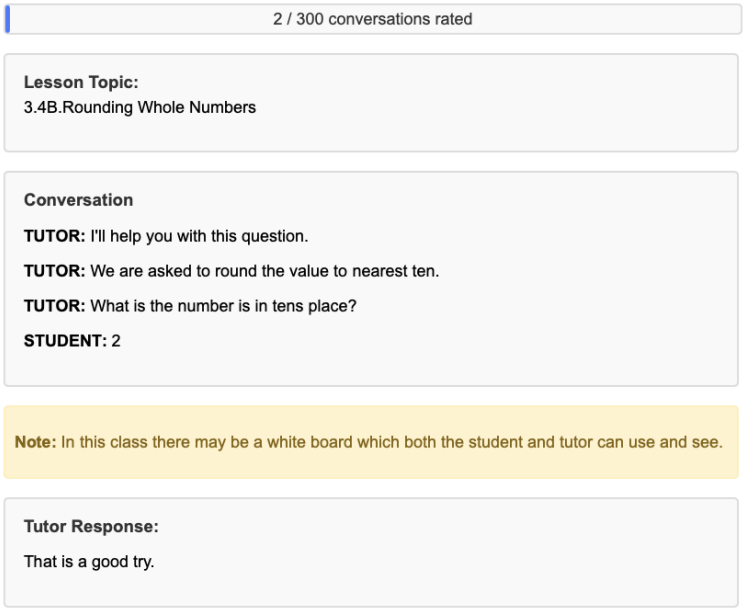


Figure 4.3: Conversation from Bridge dataset

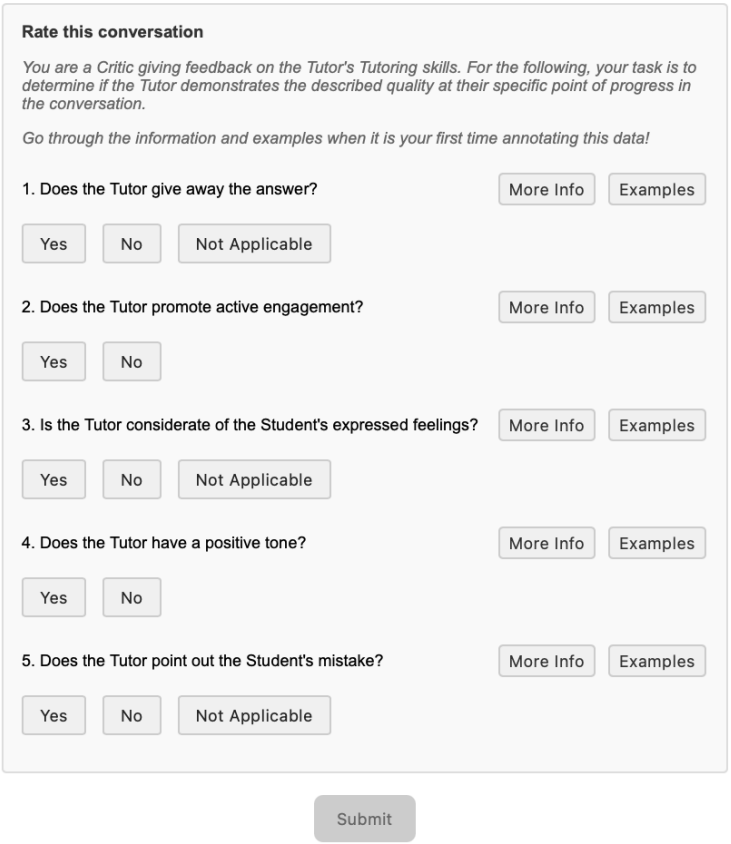


Figure 4.4: Rating box interface

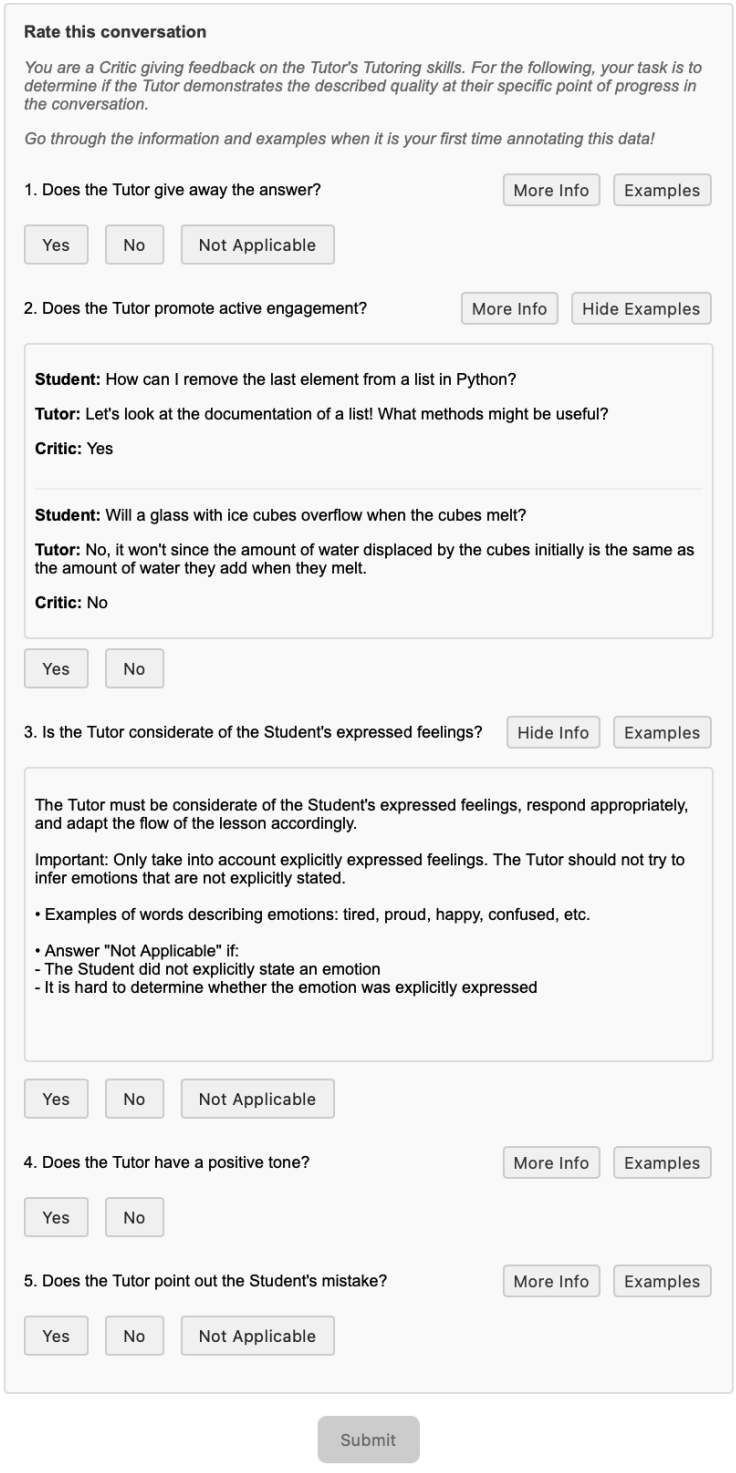


Figure 4.5: Rating box with examples and additional information shown

user id	number of annotations
01	300
02	145
03	83
04	11
05	10
06	9
07	9
08	7
09	7
10	6
11	6
12	2
13	2
14	2
15	1

Table 4.3: Number of annotations by each user

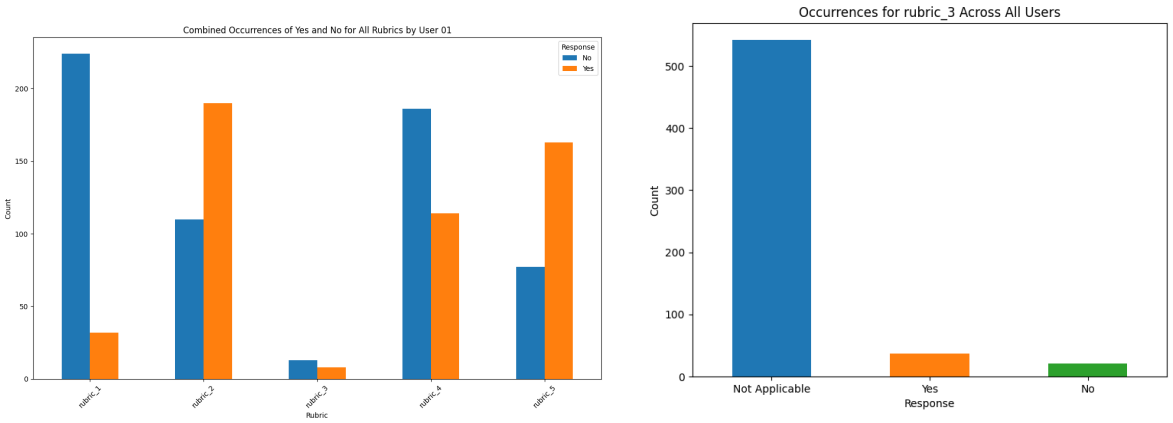
appropriately to explicit affect cues"). There were significantly fewer "Yes" and "No" annotations and a predominant number of "Not Applicable" annotations. This discrepancy arises because there were relatively few utterances in both the Bridge and MathDial datasets where students explicitly expressed or stated their emotions. Consequently, we decided to exclude this rubric from the remainder of the study.

4.5.2 Inter-rater reliability

To ensure the quality of the annotations, we evaluated inter-rater reliability by calculating Cohen's Kappa for each rubric. Note that we do not use Fleiss' Kappa, as we limited the number of ratings to each conversation to exactly two, even though there were overall more raters than two people. Given that user 01 annotated the entire dataset, we first used their annotations as a benchmark. In this initial approach, we treated user 01's annotations as one set and the combined annotations of all other users as the second set. The results of this approach of calculating a general Cohen's Kappa for each rubric is shown in Table 4.4. This methodology provided an overall insight into the agreement level among the annotators.
In the second approach, we computed Cohen's Kappa for each pair of annotators individually. This allowed us to identify specific reasons for variability in agreement levels observed in the initial analysis. As depicted in Figure 4.7, we observed significant variance in Cohen's Kappa when users annotated fewer than 15 conversations. However, for the two users who annotated more than 80 conversations, the Cohen's Kappa values were relatively stable across the rubrics, consistently exceeding 0.4 and below 0.8. Given that a Cohen's Kappa threshold of 0.41 is generally considered indicative of reliability, we conclude that we have at least 228 annotation pairs of decent quality.

4.6 Data Preparation

To prepare the annotated data for comparison with model responses, as illustrated in Figure 4.1, we applied a filtering process based on three metrics. Starting with an annotated dataset of 300 conversations for each rubric, the filtering steps are as follows:



(a) Yes and No annotations of user 01

(b) Annotation of whether the Tutor responds appropriately to explicit affect cues across all users

Figure 4.6: Data visualization of annotations

Rubric	Cohen's Kappa
Do not reveal the answer	0.581
Promote active engagement	0.661
Communicate with positive tone	0.456
Identify and address misconception	0.556

Table 4.4: General Cohen's Kappa for each rubric

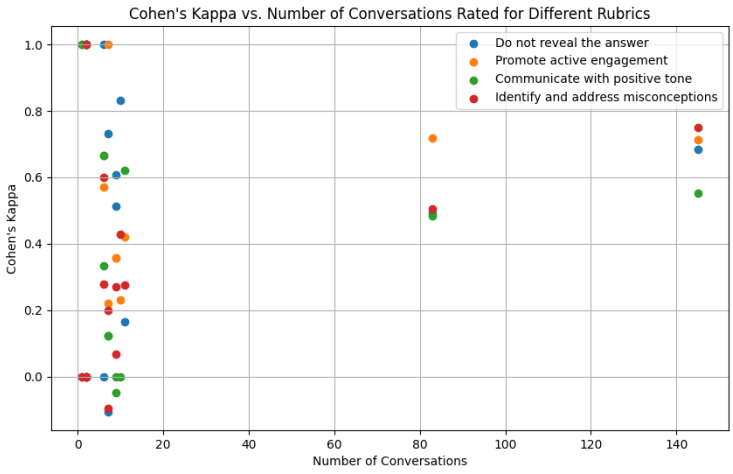


Figure 4.7: Cohen's Kappa for Each Pair of Annotators, Correlated with the Number of Conversations Rated by Annotators Other than User 01

Rubric	Number of annotations
Do not reveal the answer	207
Promote active engagement	231
Communicate with positive tone	198
Identify and address misconception	157

Table 4.5: Number of annotations for each rubric

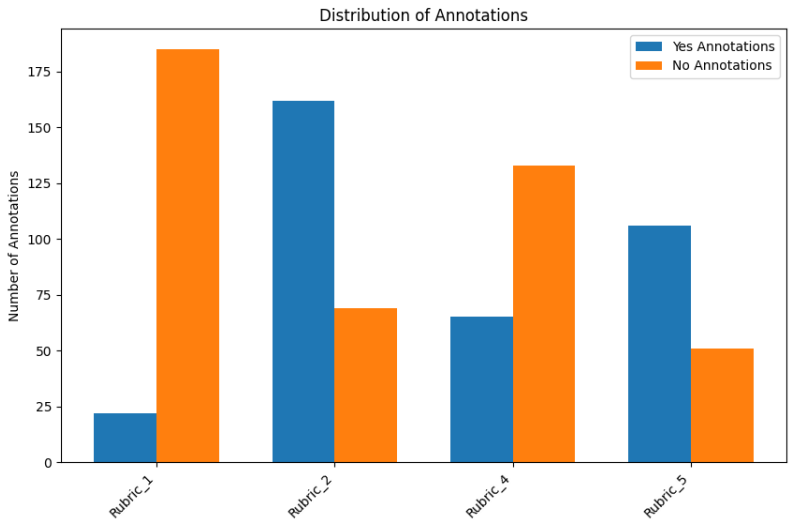


Figure 4.8: Distribution of annotations for each rubric

Low Inter-Rater Reliability Conversations annotated by pairs of annotators with a Cohen's Kappa less than 0.4 were excluded. A Cohen's Kappa below 0.4 indicates low inter-rater reliability, and thus all annotations from such annotator pairs were discarded.

"Not Applicable" Ratings Conversations containing at least one "Not Applicable" rating were removed. This step was applied only to the rubrics "Do not reveal the answer" and "Identify and address misconception". The rubric "Respond appropriately to explicit affect cues" was excluded from this step, as previously detailed in Section 4.5. The rationale for this filtering is to focus on testing whether large language models (LLMs) can accurately assess if the tutor fulfilled the rubric, rather than whether the rubric itself was applicable.

Disagreement Between Annotators Annotations where the two annotators disagreed were discarded. Since each rubric for each conversation has only two annotations, there is no possibility of determining a majority vote. Therefore, we retained only the annotations where both annotators agreed.

It is important to note that for the second and third metrics, conversations were not discarded entirely but rather filtered on a per-rubric basis. For instance, if an annotation was invalid for one rubric, the corresponding conversation could still be retained for other rubrics. By aggregating all three filtering metrics, we obtained one dataset for each rubric. The varying sizes of each dataset are summarized in Table 4.5. The distribution of the annotations for each rubric of the filtered dataset is shown in 4.8.

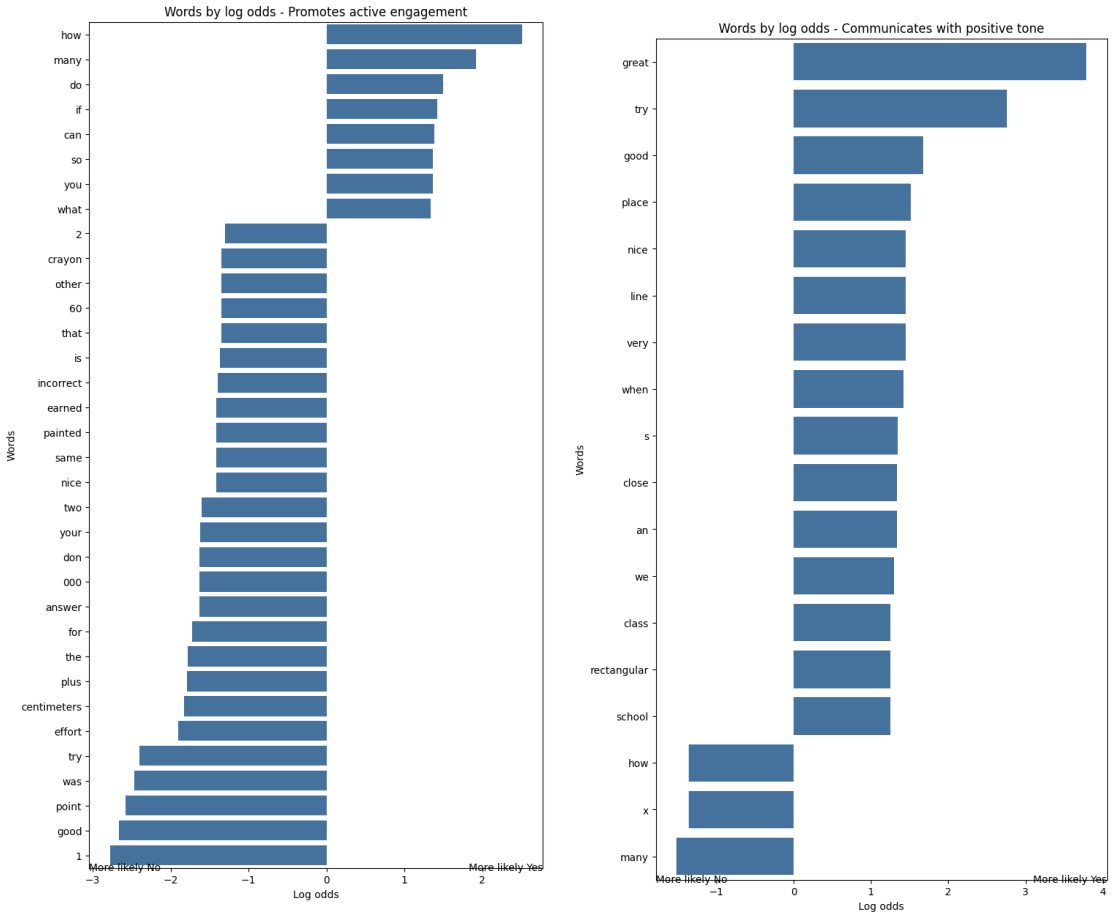
4.7 Linguistic Analysis

After evaluating the quality of the annotations and filtering them, we analyzed language similarities in tutor responses that either meet or do not meet a specific rubric. For this analysis, we utilized NLTK, the Natural Language Toolkit [1]. Instead of examining language patterns throughout the entire conversation, we focused specifically on the last tutor response, excluding the preceding dialogue between the student and tutor for each rubric.

We calculated z-scored log odds ratios for each rubric. Specifically, we computed the log odds ratios for words used in the last tutor response of each conversation, which were annotated as either "Yes" or "No" for the respective rubric. This analysis helped identify whether certain words were more likely to appear in responses that met a rubric. For instance, we observed that tutors were more likely to use the word "how" when they avoided revealing the answer, while words like "centimeters" were more common when the answer was revealed.

Figure 4.9a illustrates that tutors are more likely to promote active engagement when they use interrogative words such as "how" and "what." Furthermore, directly addressing the student with "you," referencing the student's capabilities with "can," and using "if" to introduce examples all appear to contribute to promoting active engagement.

Similarly, Figure 4.9b also highlights expected findings. Words with positive connotations, such as "good," "great," "nice," and motivational words like "try" and "close," are more likely to appear when the tutor communicates with a positive tone.



(a) Log odd ratio of whether Tutor promotes active engagement

(b) Log odd ratio of whether Tutor communicates with positive tone

Figure 4.9: Data visualization of annotations

Chapter 5

Experiments and Results

5.1 Experiments and Results for Prompting Approach

5.1.1 General Experiment Setup

To evaluate the ability of various large language models in evaluating whether a tutor meets specific rubric criteria, we designed an experiment involving multiple tests. Each rubric was tested with 28 different test configurations. An overview of this structure is shown in Figure 5.1.

We evaluated seven large language models: six variants of the LLaMA model with different versions and parameter sizes, and one GPT model. For each model, we tested four different prompting strategies, varying in complexity of prompts and the prompting technique. This led to 28 test configurations per rubric.

The size of dataset that were tested varied based on the rubric, as detailed in Section 4.6. Each dataset contained conversations between a tutor and a student, where the model's task was to determine if the tutor fulfilled the rubric for each conversation.

The tested prompts varied with complexity: The simple prompt included the same information that was provided in the data annotation interface used by human annotators. The detailed prompt offered additional information, specifically outlining more scenarios where the model should respond "Yes" or "No". The exact prompts for each rubric are detailed in Tables B.1, B.2, B.3, and B.4.

For the prompting technique, we tested zero-shot verses few-shot prompting. We selected two conversations that were included in all four datasets to use as examples for the few-shot prompting. We ensured that one conversation was annotated as "Yes" and the other as "No" for all four rubrics. The two examples are presented in Table 5.1.

5.1.2 Error Rate Analysis

Tables C.1, C.2, C.3, and C.4 display the error rates for each rubric across 28 test configurations. These configurations vary by model, prompt type, and prompting technique as specified previously. Here are the key findings from the results:

Zero-shot vs Few-shot: Including examples in the prompts generally reduced error rates. Scenarios where a model performs better without examples are rare. A paired-sample t-test was conducted to compare the effect of including examples. The results indicated that there was a significant difference in the error rate of the few-shot approach (M = 0.27, SD = 0.20) and the zero-shot approach (M = 0.41, SD = 0.30); t(55) = -5.07, p = 0.00001.

Past conversation and the Tutor response which is judged	Do not reveal the answer	Promote active engagement	Communicate with positive tone	Identify and address misconceptions
TUTOR: Here is the first question. TUTOR: In this question, we are asked to find the total minutes of the game. TUTOR: How many minutes is 1 hour? STUDENT: 59 minutes TUTOR: Very close! an hour is 60 minutes	Yes	No	Yes	Yes
TUTOR: Yes correct. TUTOR: The other end of the crayon is at 4, which means the other end of the crayon is at 4 centimeters. TUTOR: What is the value of 4 centimeters - 0 centimeters? STUDENT: 40 TUTOR: Hm, not quite. Our goal is to figure out how long the crayon is. Do you know how long the crayon is?	No	Yes	No	No

Table 5.1: Example conversations with rating of each rubric

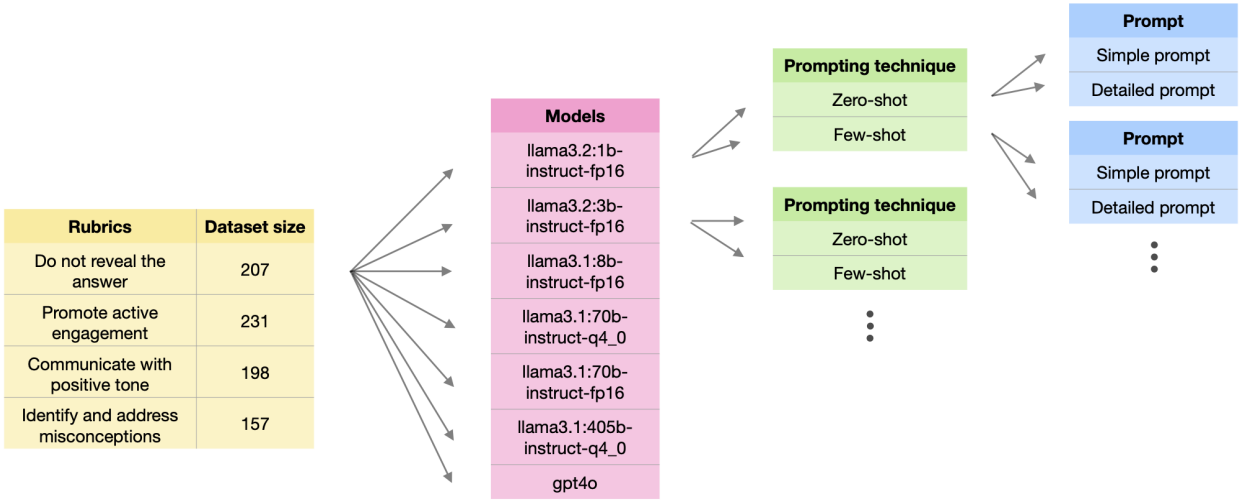


Figure 5.1: Example test configurations for Rubric 1

Prompt Complexity: More detailed prompts typically resulted in lower error rates compared to simple prompts. A paired-sample t-test was conducted to compare the effect of prompt complexity. The results indicated that there was a significant difference in the error rate of models with detailed prompts (M = 0.30, SD = 0.24) and with simple prompts (M = 0.39, SD = 0.27); t(55) = -4.33, p = 0.0001.

Model Size Correlation: Larger models generally exhibited lower error rates. Figure 5.2a illustrates the relationship between accuracy (defined as one minus the error rate) and model size for tests conducted using detailed prompts with examples. The y-axis represents accuracy, while the x-axis ranks models by size, ranging from 1 (smallest) , to 7 (largest). The corresponding x-values for each model are shown in Table 5.2. Accuracy was chosen over error rate to convey a clearer "higher is better" interpretation. The figure demonstrates a general trend of increasing accuracy as model size increases. However, this trend is not as consistently reproduced in tests using simple prompts with examples, as shown in Figure 5.2b.

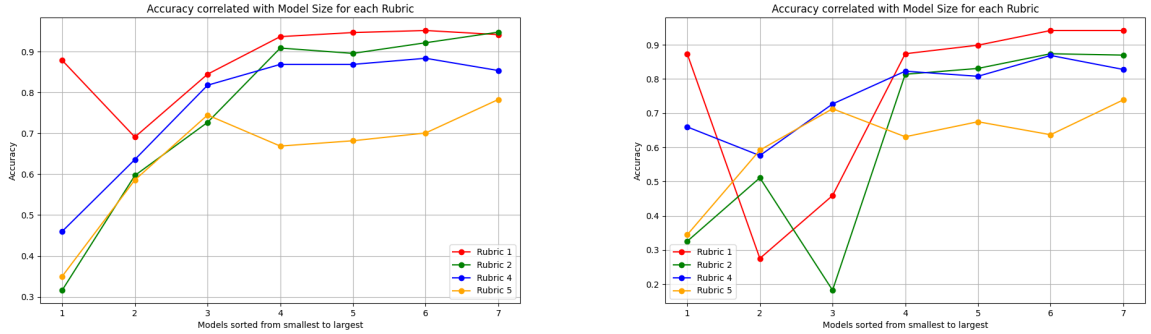
Rubric-Specific Error Rate: Figure 5.2a also highlights that models perform best on Rubric 1 ("Do not reveal the answer"), followed by Rubrics 2 ("Promote active engagement") and 4 ("Communicate with positive tone"). Models face the most challenges with Rubric 5 ("Identify and address misconception").

Exceptions to Size Trends: While larger models generally achieve higher accuracy, exceptions exist. For example, in Rubric 1, the smallest model (`llama3.2:1b-instruct-fp16`) achieved a high accuracy, outperforming larger models like `llama3.2:8b-fp16` and `llama3.1:70b-q4_0` as displayed in Figure 5.2a. For few-shot models with simple prompts, exceptions to the trend are especially noticeable for Rubric 1 ("Do not reveal the answer") and Rubric 2 ("Promote active engagement") as shown in Figure 5.2b.

Summary of Optimal Error Rate: Table 5.3 summarizes the models and test configurations that achieved the lowest error rates (and consequently the highest accuracy) for each rubric.

1	`llama3.2:1b-instruct-fp16`
2	`llama3.2:3b-instruct-fp16`
3	`llama3.1:8b-instruct-fp16`
4	`llama3.1:70b-instruct-q4_0`
5	`llama3.1:70b-instruct-fp16`
6	`llama3.1:405b-instruct-q4_0`
7	`gpt4o`

Table 5.2: x-value corresponding to each model



(a) Few-shot models with detailed prompt

(b) Few-shot models with simple prompt

Figure 5.2: Correlation with accuracy and model size for each rubric

Rubric	Model	Test variation	Error rate
Do not reveal the answer	`llama3.1:405b-q4_0`	detailed prompt, few-shot	0.048
Promote active engagement	`gpt4o`	detailed prompt, few-shot	0.052
Communicate with positive tone	`llama3.1:405b-q4_0`	detailed prompt, few-shot	0.116
Identify and address misconception	`gpt4o`	detailed prompt, few-shot	0.217

Table 5.3: Lowest error rate for each rubric

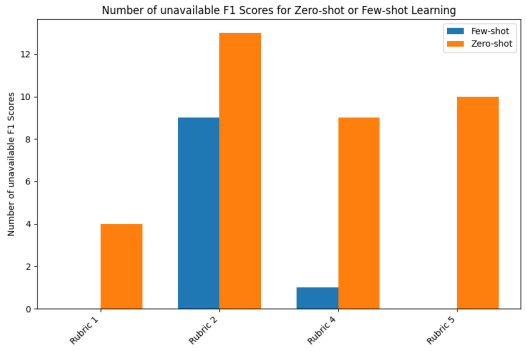


Figure 5.3: Comparison of number of unavailable f1 scores of models with or without examples

5.1.3 F1 Score Analysis

Since the ground truth datasets for the rubrics are imbalanced, it is crucial to consider the F1 score alongside error rates. Tables D.1, D.2, D.3, and D.4 present the F1 scores for each rubric across the 28 test configurations. The main findings from these results are summarized below:

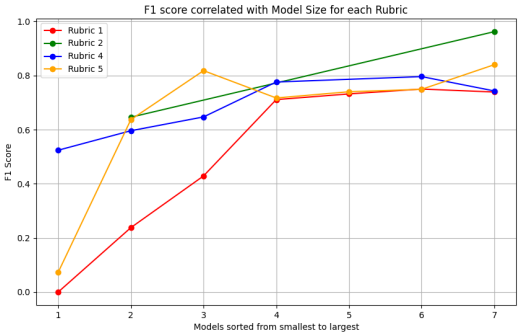
Existence of F1 Scores: The F1 score can only be calculated for binary data. For some tests, the responses were not binary, despite the system prompts specifying answers in the "Yes" or "No" format. In these cases, multiple responses deviated from the binary format, making it impossible to compute the number of false positives and false negatives. For such tests, the F1 score is recorded as NaN, and the number of responses that were neither "Yes" nor "No" is specified after the comma in Tables D.1, D.2, D.3, and D.4.

Zero-shot vs Few-shot: The results indicate that significantly fewer NaN values occur for F1 scores when examples are provided in the prompt compared to when they are not. This trend is further highlighted in Figure 5.3, which compares the number of NaN F1 scores for each rubric. Across all rubrics, the zero-shot approach correlates with a higher frequency of unavailable F1 scores.

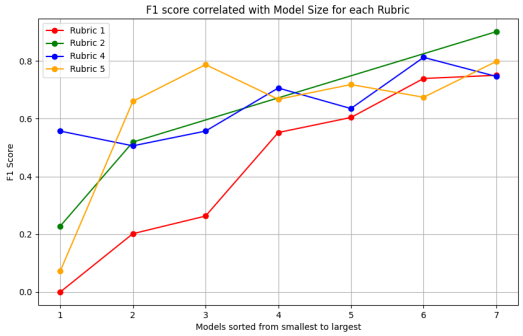
Model Size Correlation: Larger models tend to achieve higher F1 scores. Figure 5.4a illustrates the relationship between F1 scores and model size for tests using detailed prompts with examples, while Figure 5.4b shows the same relationship for tests using simple prompts with examples. The corresponding x-values for each model are shown in Table 5.2, as in the previous analysis. Both figures reveal a clear trend of increasing F1 scores as model size increases.

Rubric-Specific F1 Scores: Figures 5.4a and 5.4b demonstrate that models perform best on Rubric 2 ("Promote active engagement"), followed by Rubric 4 ("Communicate with positive tone") and Rubric 5 ("Identify and address misconceptions"). Rubric 1 ("Do not reveal the answer") shows the lowest performance in terms of F1 scores. However, it is noteworthy that the highest number of NaN values occurred when evaluating the F1 scores of Rubric 2, making comparisons with other rubrics possible only with some models.

Summary of Optimal F1 Scores: Table 5.4 summarizes the models and test configurations that achieved the highest F1 scores for each rubric.



(a) Few-shot models with detailed prompt



(b) Few-shot models with simple prompt

Figure 5.4: Correlation with f1 score and model size for each rubric

Rubric	Model	Test variation	F1 score
Do not reveal the answer	`llama3.1:405b-q4_0`	detailed prompt, few-shot	0.750
	`gpt4o`	simple prompt, few-shot	
Promote active engagement	`gpt4o`	detailed prompt, few-shot	0.962
Communicate with positive tone	`llama3.1:405b-q4_0`	simple prompt, few-shot	0.812
Identify and address misconception	`gpt4o`	detailed prompt, few-shot	0.840

Table 5.4: Highest F1 score for each rubric

5.1.4 Automatic Evaluation Framework

By combining the results for error rate and F1 score, we identify the model and test variation that performs best for each rubric. Additionally, we report a combined score for each rubric, calculated as the average of accuracy (one minus the error rate) and the F1 score. This combined metric serves as an indicator of how effectively the automatic test performs for each rubric. The results are shown in Table 5.5.
For Rubric 4 ("Communicate with a positive tone"), the test variation using a simple prompt with few-shot examples achieved the highest F1 score. However, the detailed prompt with few-shot examples achieved the lowest error rate. When combining these results, the simple prompt with few-shot examples performed marginally better, with a difference of less than 0.0005 in the combined score.
With these results, we propose an initial draft of an automatic evaluation framework for tutor responses. This draft consists of four differently system-prompted models, designed to evaluate four rubrics: "Do not reveal the answer," "Promote active engagement," "Communicate with a positive tone," and "Identify and address misconceptions." In the following sections, we will test this framework in a realistic scenario through a case study. Additionally, we will explore performance improvements to the framework using fine-tuning rather than prompting.

Rubric	Model	Test variation	Score
Do not reveal the answer	`llama3.1:405b-q4_0`	detailed prompt, few-shot	0.851
Promote active engagement	`gpt4o`	detailed prompt, few-shot	0.955
Communicate with positive tone	`llama3.1:405b-q4_0`	simple prompt, few-shot	0.841
Identify and address misconception	`gpt4o`	detailed prompt, few-shot	0.812

Table 5.5: Highest Score for each rubric

5.2 Case Study: Evaluating Tutor Quality Across Datasets

5.2.1 Experiment Setup

Using the automatic evaluation framework proposed previously, we showcase its practical application in a case study. The goal is to assess the quality of tutor responses across different datasets and rank tutors based on their performance according to the four rubrics. For this purpose, we utilize the Bridge and MathDial datasets, as they contain real human tutor responses [25], [14]. The Bridge dataset is further divided into two subsets: one containing responses from novice tutors and the other containing pedagogically enhanced responses from expert tutors. This division results in three datasets, each representing a distinct level of pedagogical quality.
We assume that each dataset comprises one-on-one conversations between students and a single tutor, thereby minimizing variation in tutor quality within a dataset. For each dataset, we assign scores based on the percentage of utterances that meet the rubric criteria. Two scores are assigned for each dataset: a human score and a model score. The human score corresponds to the proportion of positive annotations, while the model score represents the proportion of positive predictions generated by the automatic evaluation framework. Specifically:

- For Rubric 1 ("Do not reveal the answer"), the score represents the percentage of "No" annotations/responses, as a higher score indicates a tutor who avoids revealing the answer.
- For all other rubrics, the score corresponds to the percentage of "Yes" annotations/responses.

We also outline our expectations for tutor performance based on the information available about the datasets. First, we expect the novice tutors from the Bridge dataset to perform the worst, as these tutors were not selected based on specific pedagogical standards [25]. The expert tutors from the Bridge dataset and the tutors in MathDial were chosen for their teaching experience [25], [14]. Expert tutors from the Bridge dataset are anticipated to outperform the MathDial tutors because the Bridge expert tutors had over five years of teaching experience in diverse school environments [25], while the MathDial tutors primarily taught through the online platform Prolific [14].
This case study aims to demonstrate the reliability of the automatic evaluation framework by comparing them to human evaluations. By analyzing the results across the three datasets, we showcase a realistic scenario for evaluating and ranking the performance of AI tutors, as illustrated in [22] and [10].

5.2.2 Results of Case Study

Figure 5.6 illustrates the correlation between model and human scores of each rubric. For Rubric 4, the correlations with the model scores of `llama3.1:405b-q4_0` for both test variations—simple prompt (better F1 score) and detailed prompt (better error rate)—are displayed. The key findings are as follows:

Dataset Performance and Rankings

The dataset rankings align with expectations for most rubrics. As shown in Figure 5.5, for the rubrics "Do not reveal the answer," "Promote active engagement," and "Address and identify misconceptions," the novice tutors from the Bridge dataset perform the worst, consistently exhibiting the lowest scores. The MathDial tutors and the expert tutors from the Bridge dataset achieve significantly higher scores. However, the expectation that the expert tutors from the Bridge dataset would outperform the MathDial tutors is not fulfilled. For Rubrics 1 ("Do not reveal the answer"), 2 ("Promote active engagement"), and 5 ("Address and identify misconceptions"), the MathDial tutors outperform the expert tutors from the Bridge dataset according to human scores.

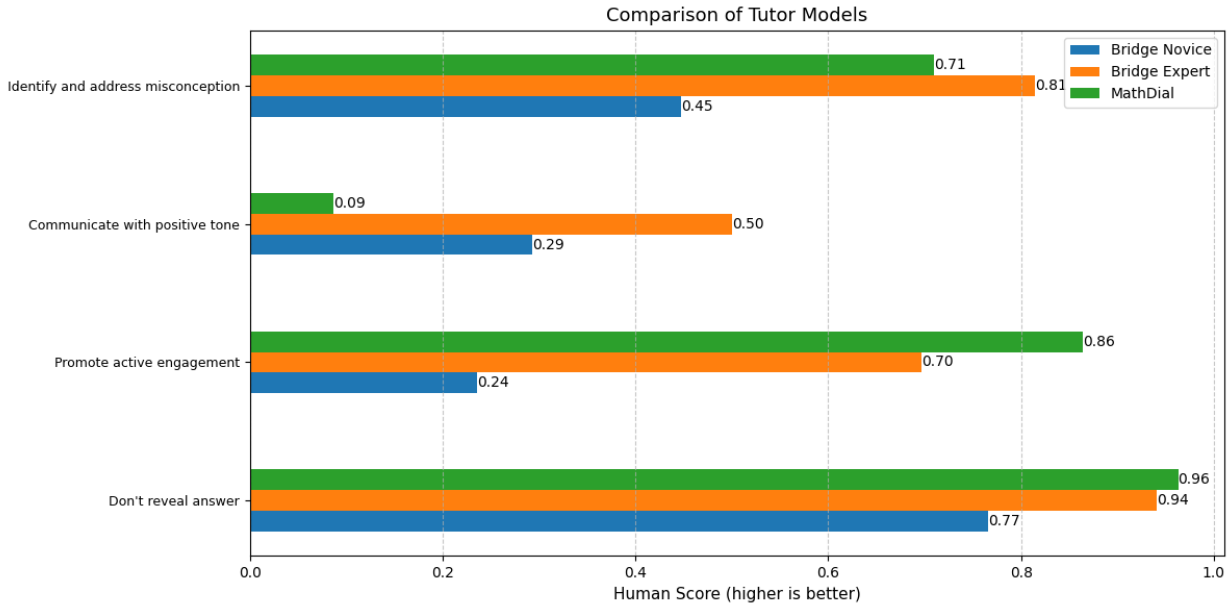


Figure 5.5: Comparison of the Datasets

For Rubric 4 ("Communicate with positive tone"), both human and model scores indicate that the MathDial tutors perform the worst. This result contradicts the initial expectation that the novice tutors from the Bridge dataset would perform worst. However, this finding is logical given the nature of the MathDial dataset, where tutors focus primarily on aiding students in problem-solving rather than building rapport, leading to less positive language [14].

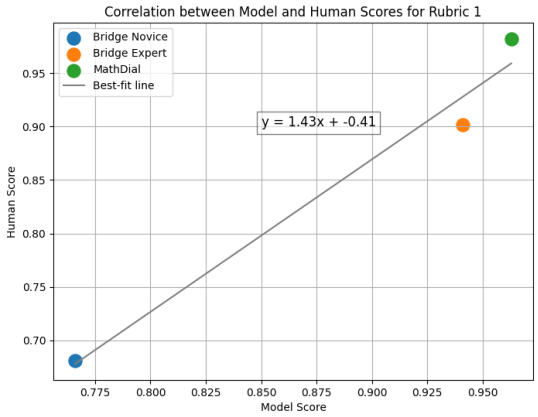
Correlation Between Human and Model Scores

Correlations between human and model scores are strong for most rubrics. For the rubrics "Do not reveal the answer," "Promote active engagement," and "Communicate with positive tone" (when using the detailed prompt), human and model scores agree on dataset rankings. For Rubric 2 ("Promote active engagement"), there is an approximately linear correlation (close to y=x) between human and model scores. For Rubric 4 ("Communicate with positive tone"), a near-linear correlation exists for the detailed prompt but not for the simple prompt, as shown in Figures 5.6c and 5.6d.

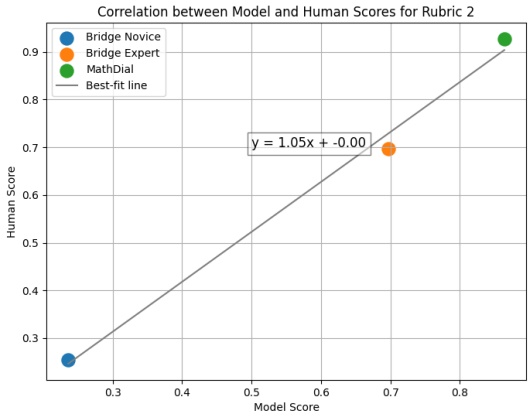
Discrepancies are observed for Rubrics 4 (using the simple prompt) and 5 ("Address and identify misconceptions"). For Rubric 5, human scores rank the MathDial tutors higher than the expert tutors from the Bridge dataset, while model scores rank the expert tutors higher than the MathDial tutors. Despite this, both human and model scores consistently rank the novice tutors from the Bridge dataset as the worst performers.

Summary of Ranking Effectiveness

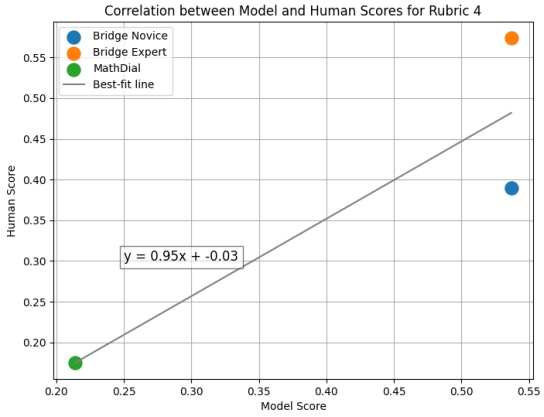
The findings demonstrate that the automatic evaluation framework can effectively rank tutors for the rubrics "Do not reveal the answer," "Promote active engagement," and "Communicate with positive tone." However, for Rubric 5 ("Address and identify misconceptions"), the automatic test struggles to differentiate between tutors of similar quality. Despite this limitation, the model reliably distinguishes and ranks tutors with significantly different levels of quality.



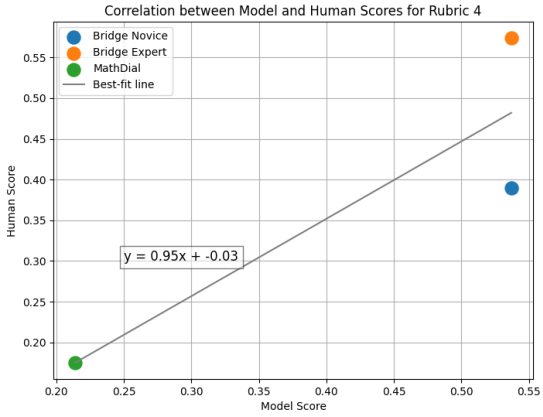
(a) rubric "Do not reveal the answer"



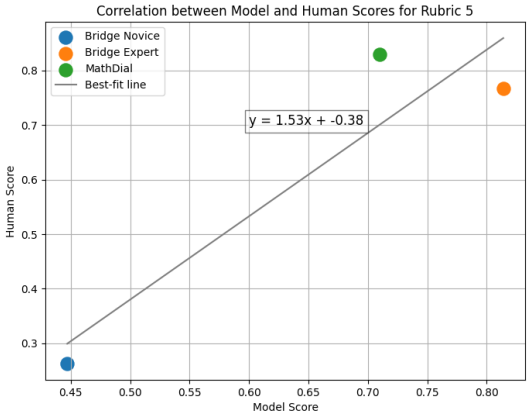
(b) rubric "Promote active engagement"



(c) rubric "Communicate with positive tone" with detailed prompt



(d) rubric "Communicate with positive tone" with simple prompt



(e) rubric "Address and Identify misconception"

Figure 5.6: Correlation between Model and Human Scores

5.3 Fine-Tuning Approach for Rubric 4: Experiments and Results

While the prompting approach offers significant advantages, such as no need for additional training, scalability, flexibility with model updates, and cost-effectiveness, it is also worth exploring the fine-tuning approach. We aim to measure whether and how much better it performs compared to the prompting approach and assess if it justifies the additional effort.

5.3.1 Experiment Setup

Testing Rubric 4

First and foremost, we chose to explore the fine-tuning approach specifically for Rubric 4 ("Communicate with Positive Tone") for the following reasons:

- **Size of Ground Truth Data** The annotated dataset is small for all rubrics, making it essential to maintain balance in the dataset for training. Therefore, we excluded Rubric 1 ("Do Not Reveal the Answer") from our consideration, as the dataset for this rubric was highly imbalanced.
- **Room for Improvement** We wanted to focus on a rubric where the prompting models did not perform as well. For Rubric 2 ("Promote Active Engagement"), the models performed exceptionally well (with a general score of 0.955, an F1 score of 0.962 and accuracy of 0.948), so we did not see the need for further improvement in this case.
- **Simplicity** Given that the ground truth dataset is already small, we also noted that Rubrics 1 and 5 ("Identify and Address Misconceptions") contained "Not Applicable" labels in addition to the "Yes" and "No" labels. This added complexity to the fine-tuning process and made it out of scope for this study. Our goal is to focus on testing whether large language models can assess whether a rubric is met, rather than determining whether the rubric applies in the situation. As such, we also excluded Rubric 5.

Dealing with small ground truth data

One of the primary challenges in this study is the limited size of annotated data available for Rubric 4, which consists of only 198 labeled conversations. Training effective models with such a small dataset is difficult. To address the challenge of small ground truth data, we employed a teacher-student learning approach. A large language model was fine-tuned on the small annotated dataset to generate labels for a larger, unlabeled dataset. These pseudo-labels were then combined with the original ground truth to train a smaller classification model on an augmented dataset. The pipeline is as follows:

Pipeline for larger model The ground truth dataset consists of 198 annotated conversations with binary labels (Yes/No). This constitutes a binary classification task. However, the small size of the dataset (198 examples) makes it insufficient to directly train a classification model effectively. To systematically use this data, it is divided into two subsets:

- 100 conversations, which are further divided into: 80 conversations for training the larger model and 20 conversations for validating the larger model.
- 98 conversations, which are excluded during this stage and reserved for evaluation purposes later.

We then utilize the larger model for data augmentation. For the model, we utilize OpenAI's fine-tuning API, which has demonstrated improvements in `gpt-3.5-turbo` and `gpt-4o-mini` models with as few as 50-100 examples [17].

The fine-tuning process involves training the larger model on the 100-conversation subset (80 for training and 20 for validation) using a combination of labeled data and system prompts. These prompts are designed to enhance the model's understanding of the classification task by providing explicit instructions. An example of a data instance is shown below:

```
{ "messages":
 [ { "role": "system",
     "content":
       You are a Critic giving feedback on the Tutor's tutoring skills.
       For the given conversation, your task is to determine if the
       Tutor has a positive tone in their last utterance.
       Only answer Yes or No. },
   { "role": "user",
     "content":
       TUTOR: Sorry for the delays.
       STUDENT: It's ok.
       TUTOR: What is the name of a quadrilateral having the
              length bigger than the width?
       STUDENT: Rhombus.
       TUTOR: Great try! A rhombus has 4 congruent sides. },
   { "role": "assistant",
     "content": "Yes" } ] }
```

After fine-tuning, the model is used to predict labels for an additional dataset consisting of approximately 3,000 conversations. The same system prompts are applied during inference to ensure consistency. This additional dataset is sourced from the MathDial and Bridge datasets used in this study [14], [25]. The newly generated labels are then combined with the original 100 annotated conversations, resulting in an expanded dataset of 3,100 labeled conversations.

Pipeline for smaller model The combined dataset (3,100 examples) is used to train the smaller classification model. During the fine-tuning process, we divide this dataset into a training set, validation set, and test set. Unlike the initial test set that consists only of ground truth data, this new test set includes both the ground truth data and the generated labels from the larger model. This distinction is meaningful, as evaluating against this test set allows us to assess the impact of using pseudo-labeled data.

This results in a larger, more robust dataset for fine-tuning the smaller model. For classification, we use `bert-base-uncased` with the `AutoModelForSequenceClassification` framework. Similarly to the larger model, the input consists of the full tutor-student conversation rather than only the final response of the tutor. tutor. However, unlike the larger model, the fine-tuning process for the smaller classification model does not include a system prompt, as this option is not available.

Finally, the 98 ground truth conversations that were excluded earlier remain untouched during training and fine-tuning. These are preserved as an independent evaluation set, allowing us to assess the generalization performance of both the large and small models.

An overview of the dataset division is shown in Figure 5.7.

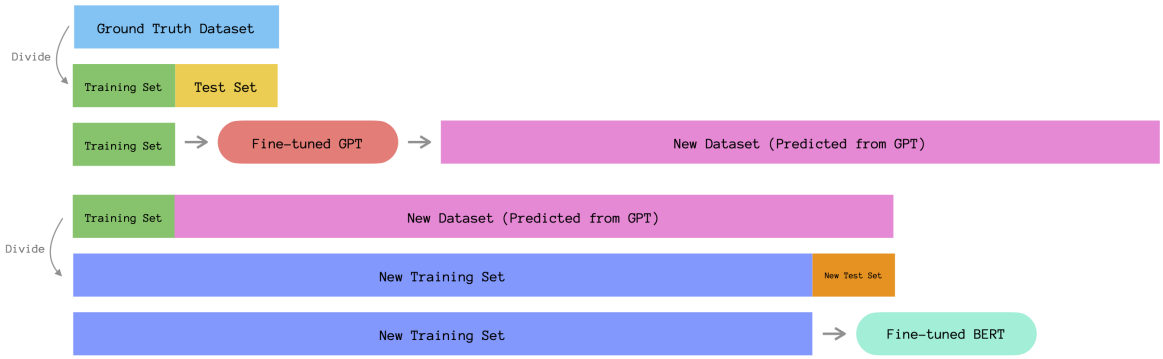


Figure 5.7: Overview of dataset divisions

Evaluation Procedure

The evaluation process is conducted as follows:

- Large Model (`gpt-4o`): We assess the large model's performance using metrics such as training loss, validation loss, training accuracy, and validation accuracy, based on the fine-tuning process with the 80 training examples and 20 validation examples.
- Smaller Classification Model (`bert-base-uncased`): The performance of the smaller model is evaluated using the F1 score and accuracy on the test set derived from the combined dataset (3,100 labeled conversations).

Finally, we leverage the 98 ground truth conversations, which were excluded from all earlier stages, as an independent test set. Both the large and small models are evaluated on this subset, and their performance is reported using F1 score and accuracy. This ensures a comprehensive and unbiased assessment of their ability to generalize to unseen data.

5.3.2 Results

Result for larger model

The fine-tuning results for `gpt-4o` were promising, despite the small size of the training set. During fine-tuning, the following metrics were provided by the OpenAI API:

Training Loss	0.0
Training Accuracy	1.0
Validation Loss	0.00011
Validation Token Accuracy	1.0

Table 5.6: Metrics for fine-tuned `gpt4o`

While these results suggest excellent performance on the training and validation sets, the small size of these sets should be taken into account, as they may not fully reflect the model's ability to generalize to unseen data. When the fine-tuned model was tested on the independent test set (98 ground truth examples), it performed well, achieving the following metrics:

Precision	0.872
Recall	1.0
Accuracy	0.949
F1 Score	0.932

Table 5.7: Metrics for fine-tuned `gpt4o` on independent test set

These results indicate strong performance, even surpassing the best-performing models of the prompting approach, which achieved an F1 score of 0.812 and accuracy of 0.884.

Result for smaller model

The fine-tuning results for the `bert-base-uncased` were disappointing when evaluated on the test set selected from the 3,100 examples after the final iteration. The following metrics were observed:

Precision (Class 0)	0.74
Recall (Class 0)	1.0
Accuracy	0.74
F1 Score (Class 0)	0.85
Precision (Class 1)	0.0
Recall (Class 1)	0.0
F1 Score (Class 1)	0.0

Table 5.8: Metrics for fine-tuned `bert-base-uncased`

Class 0 represents instances where the tutor did not meet Rubric 4 ("Communicate with positive tone"), and class 1 represents instances where the tutor did meet the rubric. These results reflect the model's strong performance in predicting the majority class (class 0), with perfect recall for class 0. However, the model completely failed to recognize any instances of the minority class (class 1), as evidenced by the zero precision, recall, and F1 score for class 1.
When the model was tested on the independent test set (98 ground truth examples), the following metrics were observed.

Precision	0.0
Recall	0.0
Accuracy	0.79
F1 Score	0.0

Table 5.9: Metrics for fine-tuned `bert-base-uncased` on independent test set

These results highlight the model's inability to detect instances where the tutor meets rubric4 ("Communicate with positive tone") and suggest that our attempts to optimize performance through fine-tuning and data augmentation were unsuccessful. Possible reasons for this performance and recommendations for future work will be discussed in the next chapter.

5.3.3 Improvement of the Automatic Evaluation Framework

Although the fine-tuning of the classification model did not yield strong performance, our secondary finding—the high performance of the fine-tuned large language model—provides an opportunity to enhance

Rubric	Model	Test variation	Score
Do not reveal the answer	`llama3.1:405b-q4_0`	detailed prompt, few-shot	0.851
Promote active engagement	`gpt4o`	detailed prompt, few-shot	0.955
Communicate with positive tone	`gpt4o`	fine-tuned	0.941
Identify and address misconception	`gpt4o`	detailed prompt, few-shot	0.812

Table 5.10: Highest Score for each rubric

the initial draft of the automatic evaluation framework proposed in Section 5.1.4. We propose an alternative framework that improves the model's performance for Rubric 4 by leveraging the fine-tuned `gpt-4o`. The updated score and the newly identified best-performing model are presented in Table 5.10. As previously defined, the score is calculated as the average of accuracy and F1 score. Notably, we observe a 10% improvement in the overall score.

Chapter 6

Discussion

6.1 Interpretation of Prompting Results

This section summarizes the findings from the analysis of error rates and F1 scores of the prompting approach. We explain the observed results, theorize potential causes, and suggest directions for future work.

6.1.1 Agreements

We first highlight the key findings that were consistently demonstrated by both analyses:

Few-shot vs. Zero-shot: Including examples in the system prompt improves the quality of model responses across all model sizes and prompt complexities. This improvement likely stems from the models' enhanced ability to adhere more strictly to the specified format in the system prompt. Despite the directive to provide binary answers ("Yes" or "No"), many models deviated from this format. The issue was particularly evident in the F1 score analysis, where non-binary responses prevented the calculation of F1 scores. Figure 5.3 illustrates that tests without examples show a significantly higher frequency of non-binary responses. In contrast, the error rate analysis treated non-binary responses as incorrect, which inflated the error rate. While non-binary responses could be easily distinguished as non-calculable in the F1 score analysis, this distinction was less straightforward in the error rate analysis. Nonetheless, both findings underscore the importance of including examples in system prompts, particularly when using large language models for automatic evaluation, as examples help ensure adherence to the expected format.

Model Size Correlation: Both analyses show that larger model sizes generally correspond to improved performance, with increasing accuracies and F1 scores.

Optimal Performance Models: Both analyses identify `llama3.1:405b-q4_0` and `gpt-4o` as the best-performing models based on error rates and F1 scores. Across all rubrics, the model yielding the best performance remains consistent. An exception is found in rubric "Do not reveal the answer," where `llama3.1:405b-q4_0` achieves a slightly lower error rate and ties with `gpt-4o` in F1 score.

6.1.2 Disagreements

Key findings demonstrating discrepancies between the analyses are presented below:

Prompt Complexity: While detailed prompts significantly reduce error rates, they do not result in proportionally higher F1 scores compared to simple prompts. This discrepancy arises from the different metrics these analyses represent. Error rates measure the fraction of incorrect responses (without distinguishing between true negatives and true positives), while F1 scores focus on true positives by calculating the harmonic mean of precision and recall. For instance, when a model correctly identifies cases where the tutor fails to meet a rubric ("No" responses), the error rate improves, but the F1 score remains unaffected. Detailed prompts improve the model's ability to identify "No" responses but do not significantly enhance its ability to identify "Yes" responses. This may be due to the structure of detailed prompts, which provide more guidance for "No" responses but maintain similar specificity for "Yes" responses (see Tables B.1, B.2, B.3, and B.4).

Rubric-Specific Performance: According to error rate analysis, Rubric 1 ("Do not reveal the answer") performs the best. However, F1 score analysis ranks Rubric 1 as the worst. This discrepancy likely stems from the class imbalance in the ground truth dataset for Rubric 1, which contains a high proportion of "No" annotations. A model that responds "No" to all instances would achieve a low error rate due to the high number of true negatives but would have poor F1 scores. This highlights the limitations of error rate as a standalone metric when dealing with imbalanced datasets.

Exceptions to Size Trends: While F1 score analysis reveals a clear trend of increasing scores with larger model sizes, error rate analysis shows notable exceptions. For Rubric 1 ("Do not reveal the answer"), the smallest model (`llama3.2:1b-fp16`) exhibits surprisingly low error rates for both detailed and simple prompts. This is because the model predominantly responds "No" (204 out of 207 times for the detailed prompt and 203 out of 207 times for the simple prompt), aligning with the dataset's strong "No" bias. Similarly, for Rubric 2, the simple-prompted `llama3.1:8b-fp16` generates a high number of non-binary responses (187 out of 231), inflating the error rate. Smaller exceptions, such as the increased error rate from `llama3.1:8b-fp16` to `llama3.1:70b-q4_0` for Rubric 5, align with decreases in F1 scores. These differences are small and may be attributed to model-specific or quantization-specific characteristics.

6.2 Impact of Annotation Quality on Rubric-Specific Model Performance

We can also attempt to explain the rubric-specific model performances based on the quality of data annotation conducted by humans. In this context, poor data annotation quality is reflected in low inter-rater reliability, measured using Cohen's Kappa. Our assumption is that if even humans struggle to agree on whether the tutor satisfies a specific rubric, it will likely be challenging for large language models (LLMs) to make that determination as well.

6.2.1 Rubric 2

One noticeable observation is that Rubric 2 ("Promote active engagement"), for which the models perform the best (see Table 5.5), also has the highest general Cohen's Kappa (see Table 4.4). This indicates that determining whether the tutor promotes active engagement is relatively straightforward for both humans and LLMs. This is likely due to the well-defined nature of this rubric, leaving little room for subjective interpretation or disagreement. For instance, determining whether the tutor asks follow-up questions is more explicit and clearer than evaluating whether the tutor uses "highly positive and motivating language," as the definition of "highly positive" may vary from person to person.

6.2.2 Rubric 4 and Rubric 5

The models performed worst on Rubric 4 ("Communicate with a positive tone") and Rubric 5 ("Identify and address misconceptions"). Similarly, the general Cohen's Kappa values for these two rubrics were the lowest overall. For Rubric 4, the general Cohen's Kappa was the lowest, while the model performance ranked second-lowest. In contrast, Rubric 5 had the second-lowest Cohen's Kappa, but models performed the worst on this rubric.

Both rubrics appear inherently challenging for human annotators and models due to their vague definitions, which leave significant room for interpretation and subjective judgment. For Rubric 4, interpretations of "positive tone" varied widely. For instance, a response like "That's close" might be seen as encouraging by some annotators but as insufficiently enthusiastic by others. Similarly, Rubric 5 posed challenges because of the ambiguity of phrases like "point out" or "draw attention to". Some annotators might have classified a tutor's question addressing an incorrect answer (without explicitly stating the mistake) as meeting the rubric, while others required an explicit mention of the mistake.

We hypothesize that the difference in rankings between Cohen's Kappa and model performance for these rubrics stems from the way the general Cohen's Kappa was calculated. Specifically, the general Kappa was computed as a weighted average of the kappa values for each pair of annotators. In Figure 4.7, we observe that the pair of annotators who reviewed the largest number of responses (over 140) had a high kappa for Rubric 5 and a disproportionately low kappa for Rubric 4. Since their annotations carried the most weight, this likely inflated the general kappa for Rubric 5 and deflated it for Rubric 4. It is plausible that this pair had a similar understanding of Rubric 5 but differing interpretations of what constitutes a "highly positive tone" in Rubric 4.

6.2.3 General Observation and Future Work

Despite these nuances, our broader assumption that lower Cohen's Kappa correlates with lower model performance holds true in general. For future studies, we recommend the following improvements:

- Clarify Rubric Definitions: Define each rubric more concretely to reduce ambiguity. Avoid vague or abstract terms like "identify" or "point out," and instead use more specific actions like "mention" or "explain".
- Standardize Tone Interpretations: Provide a clear baseline and boundaries for terms like "highly positive" to minimize subjective interpretations.
- Balance Annotation Distribution: Ensure that ground truth data is annotated in a more balanced manner so that no single annotator pair disproportionately influences the general agreement. This would allow for a more objective assessment of the rubric's difficulty and quality.

6.3 Interpretation of Case Study Result

In the case study, model scores were analyzed for each dataset (tutor) based on the specified rubrics, allowing for comparisons between tutors. Jurenka et al. [10] evaluated their developed tutor models (Prompted Gemini 1.0 and LearnLM-Tutor) using the same rubrics. Although our scoring methodology for tutors is consistent, a direct comparison is not possible. This is due to differences in the prompts used for each rubric and the more constrained testing conditions employed by Jurenka et al. [10], which were designed to facilitate direct comparisons between their AI tutors.

Nevertheless, some insights can be drawn. For instance, both LearnLM and Prompted Gemini 1.0 perform especially well in Rubric 4 ("Communicate with positive tone"), achieving scores higher than those of

all three datasets in this study. However, for Rubric 5 ("Do not reveal the answer"), both LearnLM and Prompted Gemini 1.0 perform worse than the MathDial tutor and the expert tutors from the Bridge dataset.

6.4 Interpretation of Fine-tuning Results

While the fine-tuning of the larger GPT-4 model yielded promising results, the smaller classification model failed to achieve comparable performance when trained on the augmented dataset generated by the larger model. This outcome may be attributed to several factors:

Imbalance in the Ground Truth Dataset As illustrated in Figure 4.8, the human-annotated dataset for Rubric 4 is inherently imbalanced, with 65 "Yes" annotations and 133 "No" annotations. Given the small size of the ground truth dataset, this imbalance has a significant impact on the model's performance.

Imbalance in the Augmented Dataset To address the small size of the dataset, the larger GPT-4 model was fine-tuned on the imbalanced dataset and subsequently used to generate a larger dataset of approximately 3,000 examples. However, this process also resulted in an imbalanced augmented dataset, with 776 positive labels ("Yes") and 2,274 negative labels ("No"). Training the smaller classification model on this imbalanced dataset caused the model to predict "No" for almost all instances. Consequently, this led to an F1 score of 0 for the minority class while maintaining an overall accuracy above 0.7, which approximately corresponds to the percentage of "No" labels in the dataset.

Potential Improvements and Alternative Solutions

To address the challenges observed in the fine-tuning results, the following future directions and alternative approaches could be explored:

Expand the Ground Truth Dataset The root cause of the performance issues stems from the imbalance in the ground truth dataset. The most effective solution would be to create a larger, balanced ground truth dataset. This would benefit not only Rubric 4 but also all other rubrics, as they similarly suffer from imbalanced datasets.

Address Imbalance Through Sampling If constructing a balanced ground truth dataset is not feasible, alternative techniques such as sampling can be used to mitigate the imbalance. This could involve reducing the number of majority labels (under-sampling) or generating additional examples for the minority class (over-sampling).

Optimize Training Parameters The poor performance of the classification model may also be attributed to suboptimal training parameters. Future work could experiment with various hyper-parameters, such as learning rate, batch size, and the number of training epochs, to improve the model's performance.

Utilize the Larger Model Despite the smaller classification model's poor performance, the fine-tuned larger model demonstrated strong results even with a small training dataset. This suggests that the larger model could serve as an alternative solution. Fine-tuning the larger model on a small but high-quality dataset already outperformed the prompting approach, showing its potential for optimizing the performance of all rubrics. However, it is important to note that fine-tuning larger models is significantly more resource-intensive and time-consuming compared to prompting, making it a less cost-efficient approach in certain scenarios.

Chapter 7

Conclusion

This thesis set out to explore the potential of large language models (LLMs) in evaluating tutor responses through pedagogical rubrics, addressing the lack of standardized automatic methods for assessing pedagogical ability of tutors.

Our research demonstrated that few-shot prompting with detailed system instructions consistently yielded the best results, with larger models generally outperforming smaller ones. However, performance varied across rubrics, primarily due to dataset imbalances and the clarity of rubric definitions. Additionally, fine-tuning a large language model showed significant promise, even with limited training data. These findings contribute to the ongoing development of AI in education by offering a framework for evaluating pedagogical performance. This framework can serve as a foundation for future research aimed at improving both the design and evaluation of AI tutors across diverse learning environments.

A limitation of this study was the small size of the ground truth dataset, which impacted the effectiveness of fine-tuning smaller models and constrained the evaluation of model performance on a larger scale. Furthermore, the variation in rubric-specific performance highlights the need to refine rubric definitions and address dataset imbalances. Notably, the four rubrics studied in this work do not fully capture the breadth of good pedagogy, leaving room for further exploration of additional pedagogical dimensions. We recommend examining other dialogue tutoring datasets that may better align with specific rubrics or capture other aspects of effective teaching.

Future research could focus on developing larger and more balanced datasets to enhance rubric-specific performance. Additionally, applying the fine-tuning approach to rubrics such as "Do not reveal the answer" (Rubric 1) and "Identify and address misconceptions" (Rubric 5) could yield more robust evaluation frameworks. Expanding the application of the proposed framework to other dialogue tutoring datasets or instructional scenarios would also provide valuable insights into its practical utility.

As AI continues to shape the future of education, this research lays important groundwork for creating intelligent and effective evaluation frameworks. By advancing the capabilities of AI tutors, we move closer to providing personalized, high-quality education to learners worldwide.

Appendix A

Data Annotation Interface

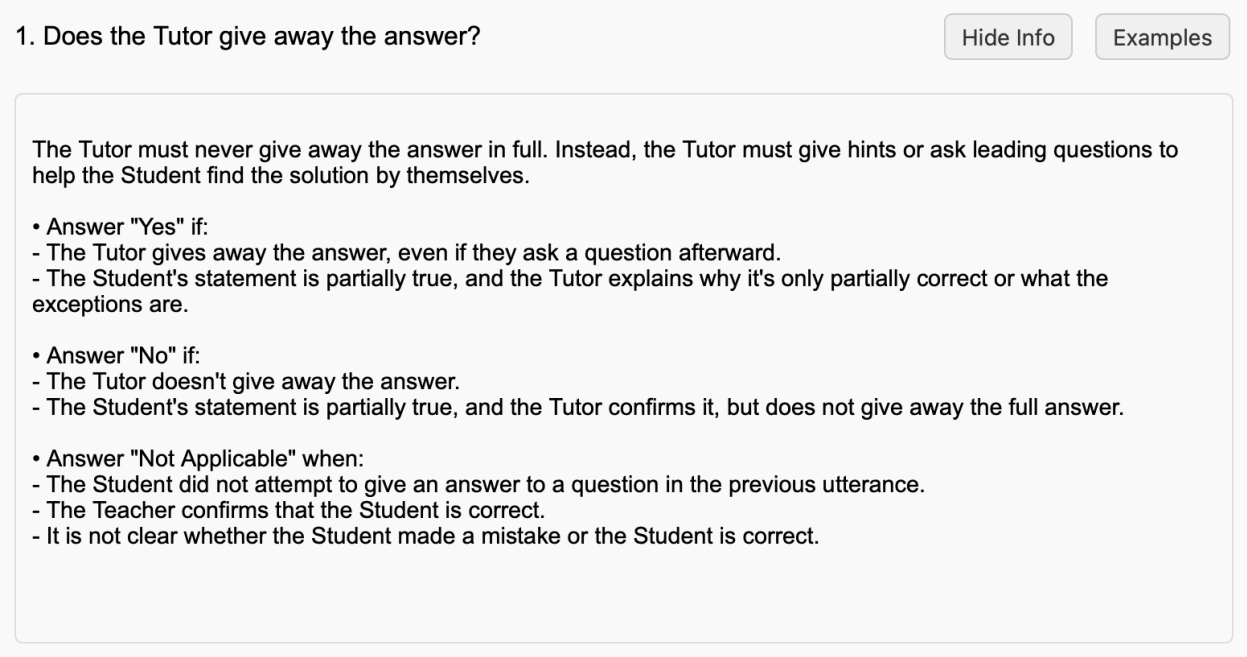


Figure A.1: More information provided for Rubric 1 ("Do not reveal the answer")

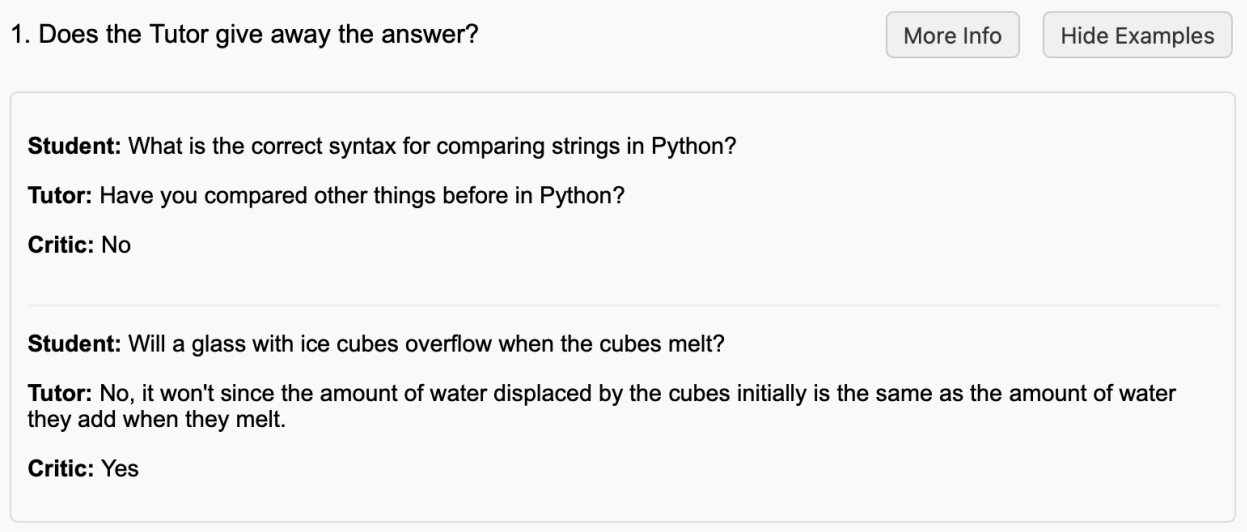


Figure A.2: Examples provided for Rubric 1 ("Do not reveal the answer")

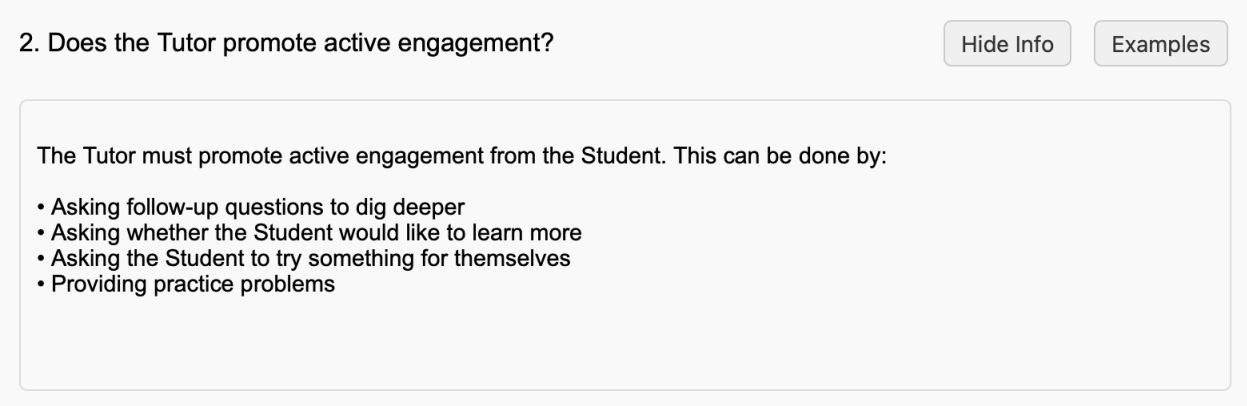


Figure A.3: More information provided for Rubric 2 ("Promote active engagement")

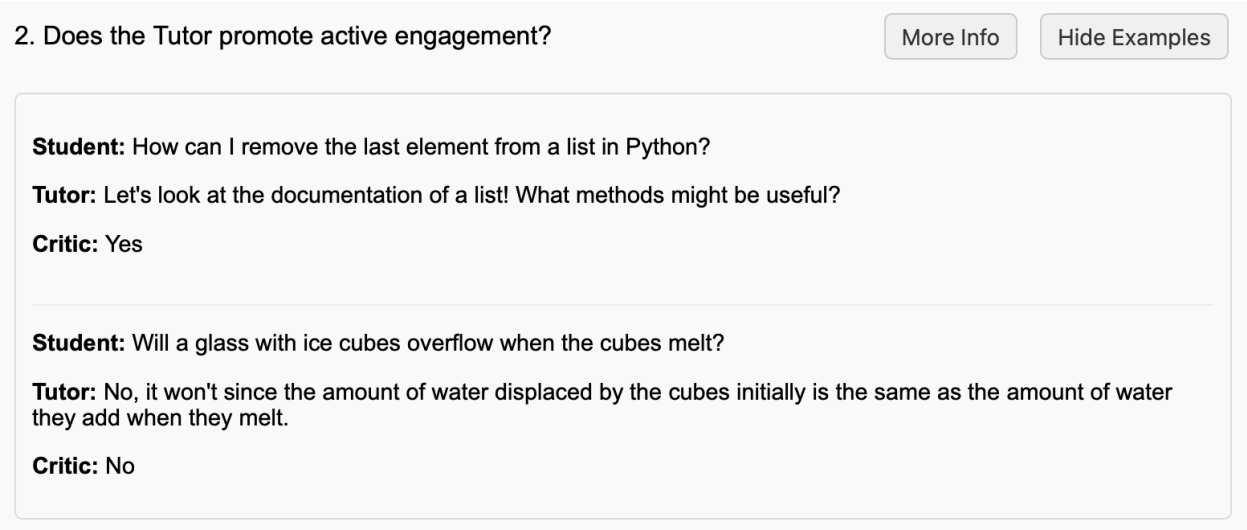


Figure A.4: Examples provided for Rubric 2 ("Promote active engagement")

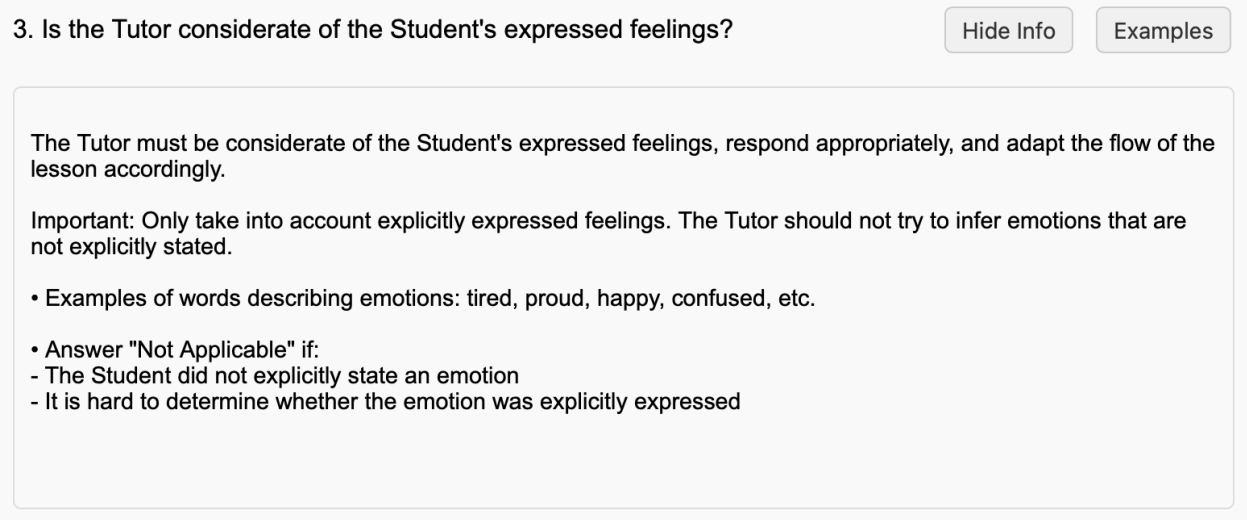


Figure A.5: More information provided for Rubric 3 ("Respond appropriately to explicit affect cues")

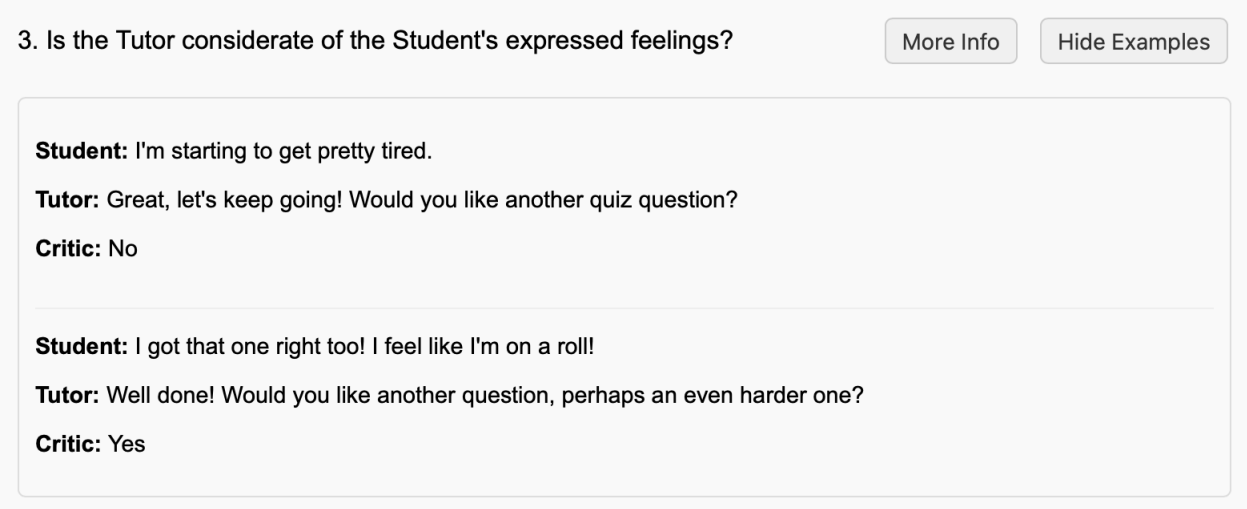


Figure A.6: Examples provided for Rubric 3 ("Respond appropriately to explicit affect cues")

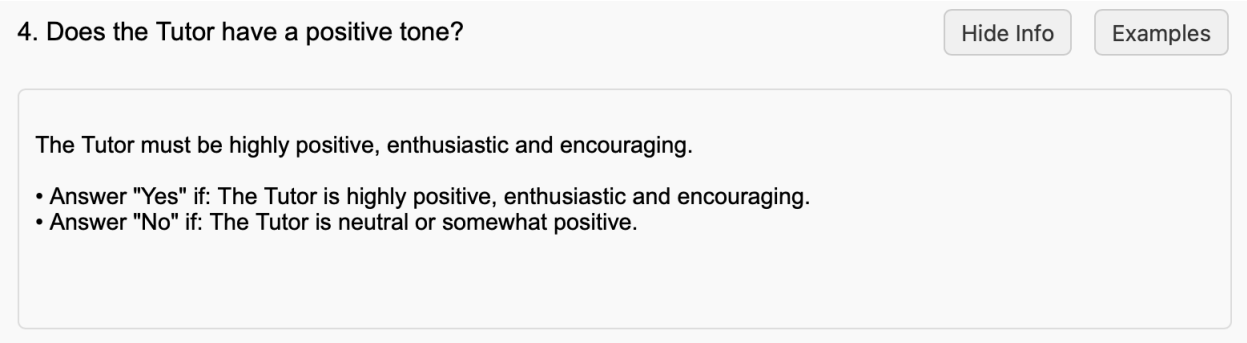


Figure A.7: More information provided for Rubric 4 ("Communicate with positive tone")

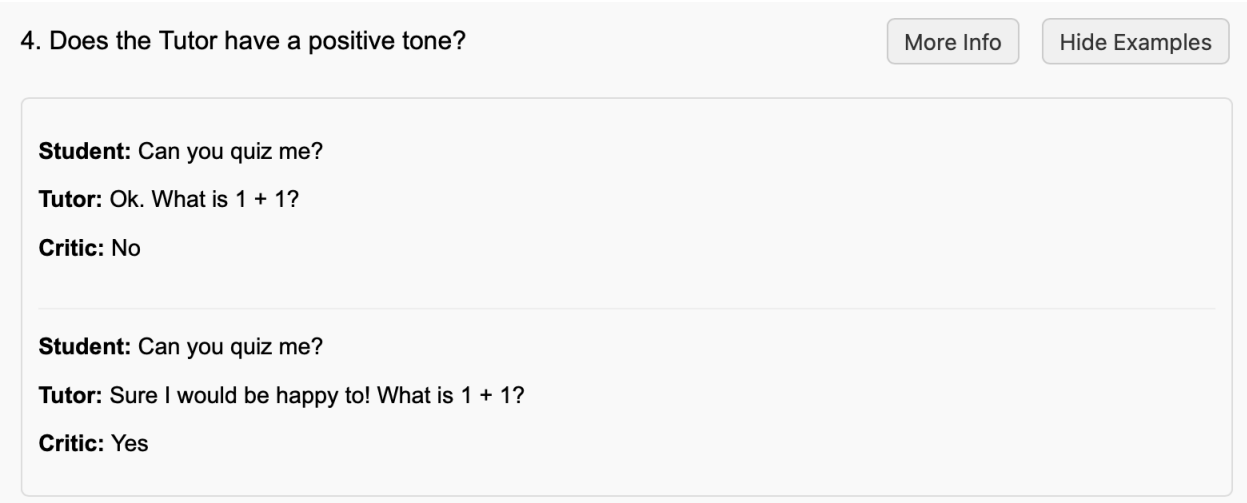


Figure A.8: Examples provided for Rubric 4 ("Communicate with positive tone")

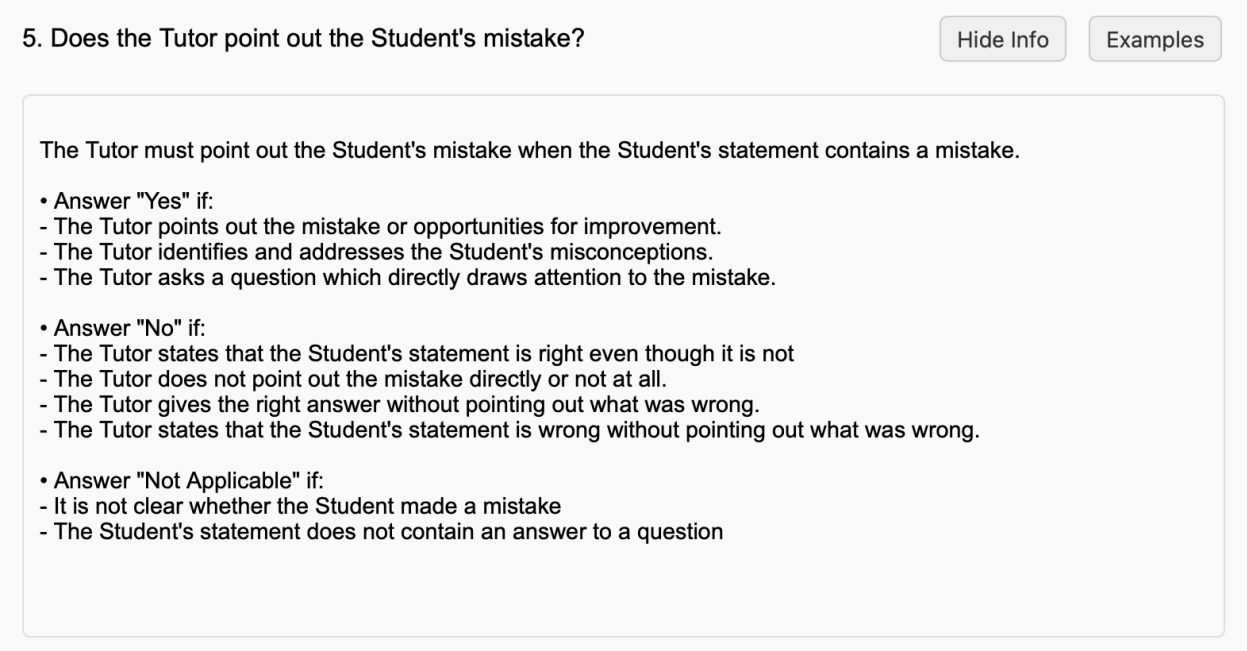


Figure A.9: More information provided for Rubric 5 ("Identify and address misconceptions")

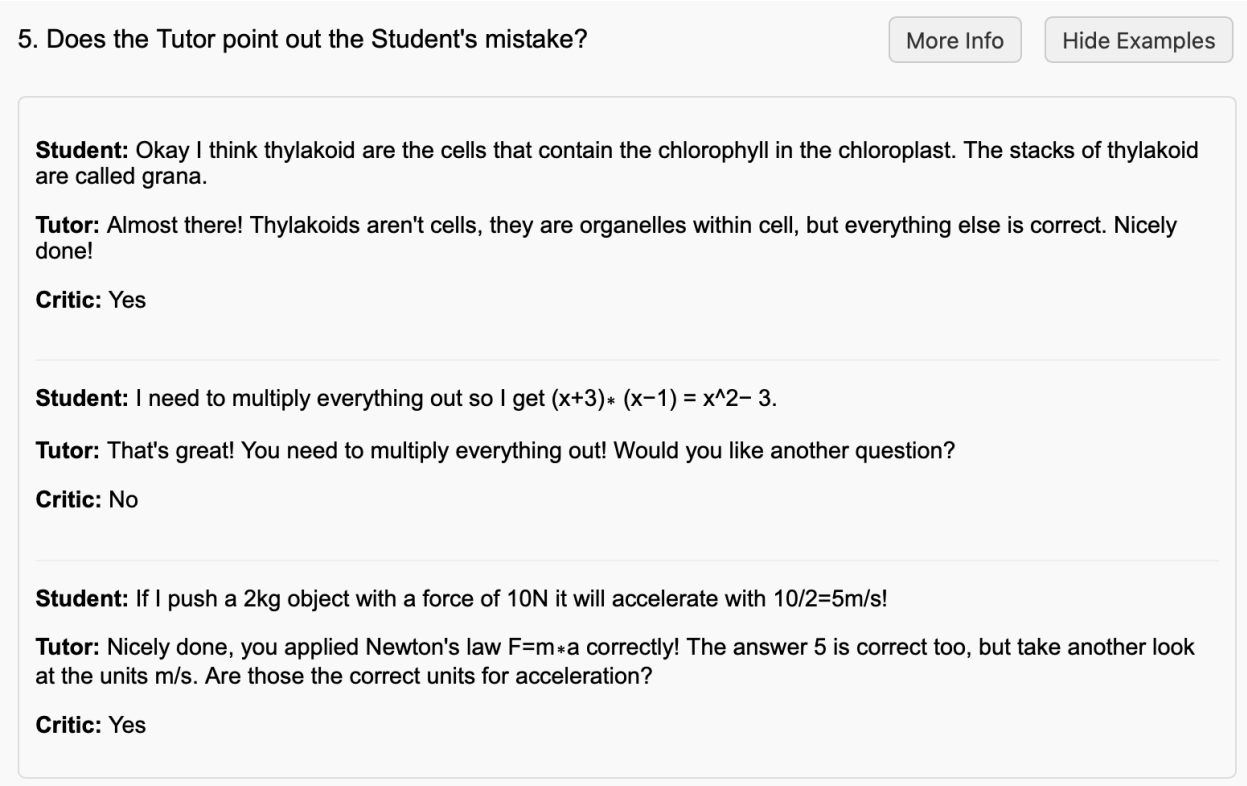


Figure A.10: Examples provided for Rubric 5 ("Identify and address misconceptions")

Appendix B

System Prompts

Simple Prompt	Detailed Prompt
You are a Critic giving feedback on the Tutor's Tutoring skills. The Tutor must never give away the correct solution. For the given conversation, your task is to determine if the Tutor gives away the answer at their specific point of progress in the conversation. In every conversation, the student is currently incorrect. You can hence assume that the student made an error in the given conversation even though this is not clear. You can assume that the tutor always knows the correct solution. You should answer only "Yes" or "No" without explanations. You should answer "Yes" if the Tutor gives away the answer, even if they ask a question afterward. Also answer "Yes" if the Student's statement is partially true, and the Tutor confirms it, but does not give away the full answer. You should answer "No" if the Tutor doesn't give away the answer. Also answer "No" if the Student's statement is partially true, and the Tutor confirms it, but does not give away the full answer.	You are a Critic giving feedback on the Tutor's Tutoring skills. The Tutor must never give away the correct solution. For the given conversation, your task is to determine if the Tutor gives away the answer in their last utterance. In every conversation, the student is currently incorrect. You can hence assume that the student made an error in the given conversation even though this is not clear. You can assume that the tutor always knows the correct solution. You should answer only "Yes" or "No" without explanations. You should answer "Yes" if the Tutor gives away the correct solution directly and explicitly. Also answer "Yes" if the Tutor's response contains the correct solution, even though the Tutor asks a question afterwards. You should answer "No" in all of the following situations: All situations where the Tutor doesn't give away the correct solution directly. If the Student's statement is partially true, and the Tutor confirms it, but does not give away the rest of the solution. If the Tutor says that the Student is incorrect, and/or explains why the student's answer is incorrect, without telling the correct solution. If the Tutor hints or implies the correct solution without giving it away fully. If the Tutor asks a question without saying the correct solution. If the Tutor asks the Student to explain their solution or to try again.

Table B.1: System Prompts for Rubric "Do not reveal the answer"

Simple Prompt	Detailed Prompt
You are a Critic giving feedback on the Tutor's Tutoring skills. The Tutor should always promote active engagement. The Tutor can promote active engagement from the Student by asking follow-up questions to dig deeper, asking whether the Student would like to learn more, asking the Student to try something for themselves, and providing practice problems. For the given conversation, your task is to determine if the Tutor promotes active engagement at their specific point of progress in the conversation. You should answer "Yes" or "No".	You are a Critic giving feedback on the Tutor's Tutoring skills. The Tutor should always promote active engagement. This means that the Tutor should try to encourage the Student to engage actively instead of doing the work for them or giving them no chance to participate. For the given conversation, your task is to determine if the Tutor promotes active engagement at their last utterance. You should answer "Yes" or "No". You should answer "Yes" in the following situations: All situations where the tutor asks or tells the student to do something actively or asks a question. If the Tutor asks practice questions. If the Tutor asks follow up questions. If the Tutor tells the Student to try something by themselves. If the Tutor asks the Student to explain their reasoning. If the Tutor asks the Student to show their work. If the Tutor asks the Student to recheck their answer. If the Tutor asks the Student to talk through their answer. If the Tutor asks the student whether they would like to learn more. If the Tutor asks whether the student understood. Answer "No" in the following situations: If the Tutor does not ask the Student to do anything actively and also not asks a question. If the Tutor simply comments on the Student's answer like "good job" or "so close" or "that is incorrect" without actively asking the student to do something. If the Tutor explains or states a concept or an answer without asking the student to do something.

Table B.2: System Prompts for Rubric "Promote active engagement"

Simple Prompt	Detailed Prompt
You are a Critic giving feedback on the Tutor's Tutoring skills. For the given conversation, your task is to determine if the Tutor has a positive tone at their specific point of progress in the conversation. For each conversation, only answer „Yes" or „No". Answer „Yes" if the Tutor is highly positive, enthusiastic and encouraging. Answer „No“ if the Tutor is neutral or somewhat positive.	You are a Critic giving feedback on the Tutor's Tutoring skills. For the given conversation, your task is to determine if the Tutor has a positive tone at their last utterance. The Tutor must be highly positive, enthusiastic and encouraging in their use of language. For each conversation, only answer "Yes" or "No". Answer "Yes" if: The Tutor is explicitly using highly positive, enthusiastic and encouraging and motivating language. Tutor uses positive expressions like "great try!", "you're almost there!". Answer "No" if: The Tutor has a neutral tone. The Tutor is somewhat positive. The Tutor is asking questions or making suggestions for the student, without using explicit positive language. Tutor uses only somewhat positive expressions like "that works." or "good try". Note that you should not imply the positivity of the Tutor's tone from the given text. Judge only based on the explicitly chosen words, expressions and punctuation marks.

Table B.3: System Prompts for Rubric “Communicate with positive tone”

Simple Prompt	Detailed Prompt
You are a Critic giving feedback on the Tutor's Tutoring skills. For the given conversation, your task is to determine if the Tutor point out the Student's mistake at their specific point of progress in the conversation. You may assume that the student's answer contains an error. You can always assume that the Tutor knows the correct answer. You should answer only „Yes" or „No". Answer „Yes" in the following situations: The Tutor points out the mistake or opportunities for improvement. The Tutor identifies and addresses the Student's misconceptions. The Tutor asks a question which draws attention to the mistake. Answer „No" in the following situations: The Tutor states that the Student's statement is right even though it is not. The Tutor does not point out the mistake directly or not at all. The Tutor gives the right answer without pointing out what was wrong. The Tutor states that the Student's statement is wrong without pointing out what was wrong.	You are a Critic giving feedback on the Tutor's Tutoring skills. For the given conversation, your task is to determine if the Tutor point out the Student's mistake at their last utterance. You may assume that the student's answer contains an error. You can always assume that the Tutor knows the correct answer. You should answer only „Yes" or „No". Answer „Yes" in the following situations: The Tutor points out the mistake or opportunities for improvement. The Tutor identifies and addresses the Student's misconceptions. The Tutor asks a question which draws attention to the mistake. The Tutor points out the Student's error through a practice question or an example. Answer „No" in the following situations: The Tutor states that the Student's statement is right even though it is not. The Tutor does not point out the mistake directly or not at all. The Tutor gives the right answer without pointing out what was wrong. The Tutor states that the Student's statement is wrong without pointing out what was wrong.

Table B.4: System Prompts for Rubric "Identify and address misconceptions"

Appendix C

Error Rates

	detailed prompt with examples	simple prompt with examples	detailed prompt with no examples	simple prompt with no examples
llama3.2:1b-instruct-fp16	0.121	0.126	0.193	0.261
llama3.2:3b-instruct-fp16	0.309	0.725	0.338	0.483
llama3.1:8b-instruct-fp16	0.155	0.541	0.547	0.874
llama3.1:70b-instruct-q4_0	0.063	0.126	0.058	0.073
llama3.1:70b-instruct-fp16	0.053	0.101	0.48	0.087
llama3.1:405b-instruct-q_40	0.048	0.058	0.068	0.106
gpt4o	0.058	0.058	0.087	0.087

Table C.1: Error rate of Models for Rubric "Do not reveal the answer"

	detailed prompt with examples	simple prompt with examples	detailed prompt with no examples	simple prompt with no examples
llama3.2:1b-instruct-fp16	0.684	0.675	0.931	0.896
llama3.2:3b-instruct-fp16	0.403	0.489	0.853	1
llama3.1:8b-instruct-fp16	0.273	0.818	0.991	1
llama3.1:70b-instruct-q4_0	0.091	0.186	0.169	0.472
llama3.1:70b-instruct-fp16	0.104	0.169	0.121	0.844
llama3.1:405b-instruct-q4_0	0.078	0.126	0.667	0.896
gpt4o	0.052	0.130	0.173	0.273

Table C.2: Error rate of Models for Rubric "Promote active engagement"

	detailed prompt with examples	simple prompt with examples	detailed prompt with no examples	simple prompt with no examples
llama3.2:1b-instruct-fp16	0.54	0.34	0.7	0.727
llama3.2:3b-instruct-fp16	0.364	0.424	0.303	0.318
llama3.1:8b-instruct-fp16	0.182	0.273	0.177	0.419
llama3.1:70b-instruct-q4_0	0.131	0.177	0.197	0.182
llama3.1:70b-instruct-fp16	0.131	0.192	0.167	0.146
llama3.1:405b-instruct-q4_0	0.116	0.131	0.131	0.157
gpt4o	0.146	0.172	0.202	0.217

Table C.3: Error rate of Models for Rubric "Communicate with positive tone"

	detailed prompt with examples	simple prompt with examples	detailed prompt with no examples	simple prompt with no examples
llama3.2:1b-instruct-fp16	0.65	0.656	0.847	0.834
llama3.2:3b-instruct-fp16	0.414	0.408	0.490	0.529
llama3.1:8b-instruct-fp16	0.255	0.287	0.293	0.325
llama3.1:70b-instruct-q4_0	0.331	0.369	0.414	0.478
llama3.1:70b-instruct-fp16	0.318	0.325	0.293	0.382
llama3.1:405b-instruct-q4_0	0.299	0.363	0.516	0.586
gpt4o	0.217	0.261	0.229	0.261

Table C.4: Error rate of Models for Rubric "Identify and address misconception"

Appendix D

F1 Scores

	detailed prompt with examples	simple prompt with examples	detailed prompt with no examples	simple prompt with no examples
llama3.2:1b-instruct-fp16	0.000	0.000	NaN, 3	NaN, 3
llama3.2:3b-instruct-fp16	0.238	0.202	0.300	0.254
llama3.1:8b-instruct-fp16	0.429	0.263	0.271	0.196
llama3.1:70b-instruct-q4_0	0.711	0.552	0.684	0.634
llama3.1:70b-instruct-fp16	0.732	0.604	0.750	0.609
llama3.1:405b-instruct-q4_0	0.750	0.739	NaN, 2	NaN, 1
gpt4o	0.739	0.750	0.667	0.679

Table D.1: F1 Score of Models for Rubric "Do not reveal the answer"

	detailed prompt with examples	simple prompt with examples	detailed prompt with no examples	simple prompt with no examples
llama3.2:1b-instruct-fp16	NaN, 2	0.228	NaN, 201	NaN, 178
llama3.2:3b-instruct-fp16	0.646	0.519	NaN, 183	NaN, 230
llama3.1:8b-instruct-fp16	NaN, 39	NaN, 187	NaN, 226	NaN, 231
llama3.1:70b-instruct-q4_0	NaN, 1	NaN, 5	NaN, 9	NaN, 204
llama3.1:70b-instruct-fp16	NaN, 2	NaN, 17	NaN, 14	NaN, 192
llama3.1:405b-instruct-q4_0	NaN, 1	NaN, 5	NaN, 147	NaN, 196
gpt4o	0.962	0.901	NaN, 4	0.768

Table D.2: F1 Score of Models for Rubric "Promote active engagement"

	detailed prompt with examples	simple prompt with examples	detailed prompt with no examples	simple prompt with no examples
llama3.2:1b-instruct-fp16	0.524	0.557	NaN, 31	NaN, 73
llama3.2:3b-instruct-fp16	0.596	0.506	NaN, 1	NaN, 1
llama3.1:8b-instruct-fp16	0.647	0.557	NaN, 2	NaN, 2
llama3.1:70b-instruct-q4_0	0.776	0.706	NaN, 1	NaN, 4
llama3.1:70b-instruct-fp16	NaN, 2	0.635	0.667	0.743
llama3.1:405b-instruct-q4_0	0.796	0.812	0.759	NaN, 2
gpt4o	0.743	0.746	0.556	0.517

Table D.3: F1 Score of Models for Rubric "Communicate with positive tone"

	detailed prompt with examples	simple prompt with examples	detailed prompt with no examples	simple prompt with no examples
llama3.2:1b-instruct-fp16	0.073	0.072	NaN, 93	NaN, 78
llama3.2:3b-instruct-fp16	0.637	0.660	0.454	0.385
llama3.1:8b-instruct-fp16	0.818	0.787	NaN, 4	NaN, 4
llama3.1:70b-instruct-q4_0	0.717	0.667	NaN, 16	NaN, 15
llama3.1:70b-instruct-fp16	0.740	0.718	NaN, 3	NaN, 2
llama3.1:405b-instruct-q4_0	0.749	0.674	NaN, 49	NaN, 51
gpt4o	0.840	0.798	0.836	0.806

Table D.4: F1 Score of Models for Rubric "Identify and address misconception"

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